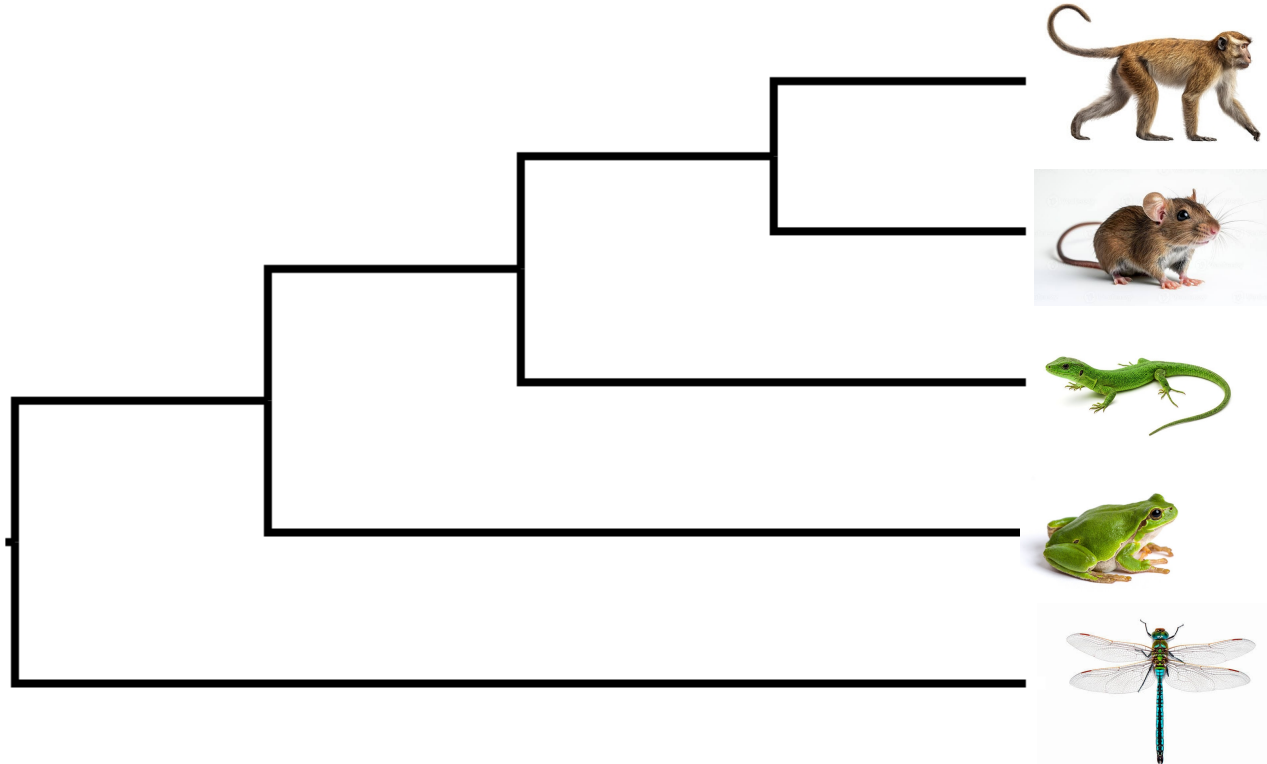


Workshop: Methods for inferring gene flow from two types of molecular data:
population genomic and phylogenetic data

Inferring gene flow from phylogenetic data by
using Aphid

What is phylogenetics?

"Phylogenetics is the study of evolutionary relationships among biological entities"



Phylogenetics using molecular data

The evolutionary histories between biological entities are mainly studied using molecular sequences, because:

- DNA is replicated and vertically transmitted over the generations.

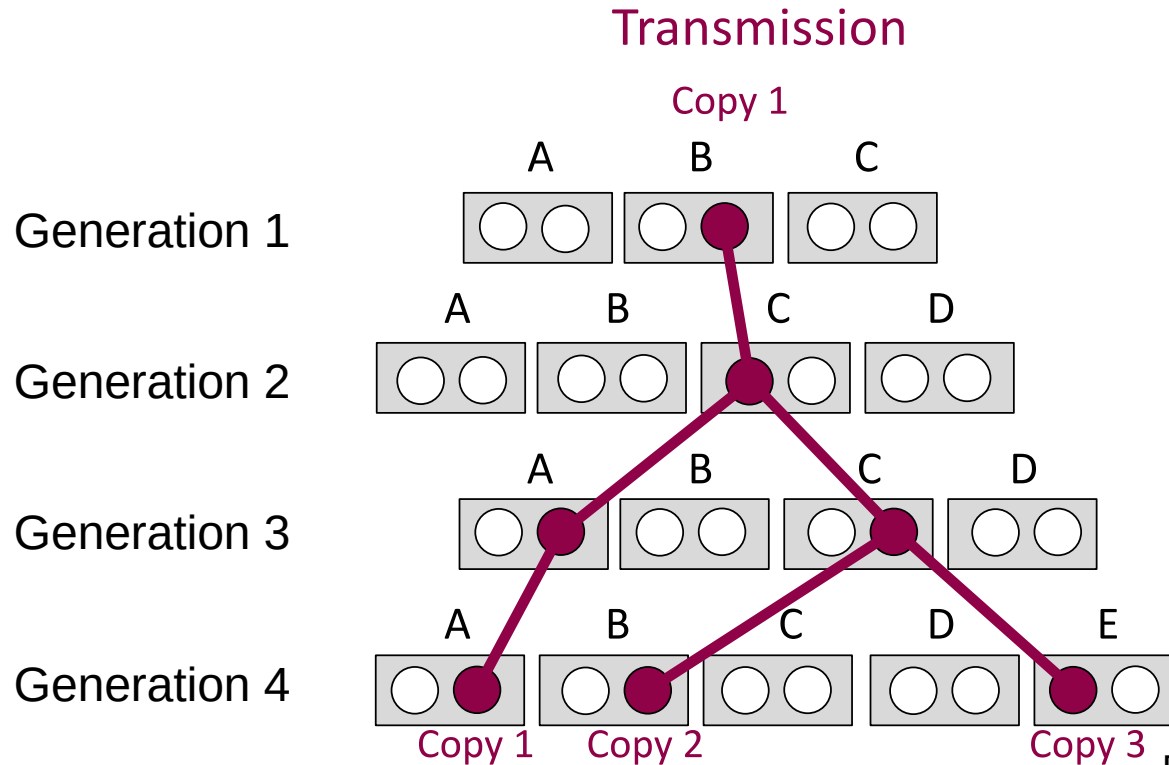


Figure from a course made by **Celine Poux**

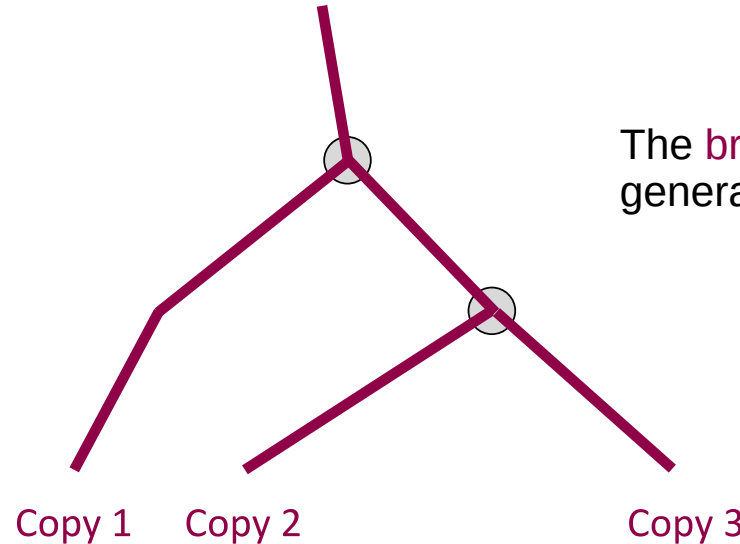
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Transmission

Copy 1

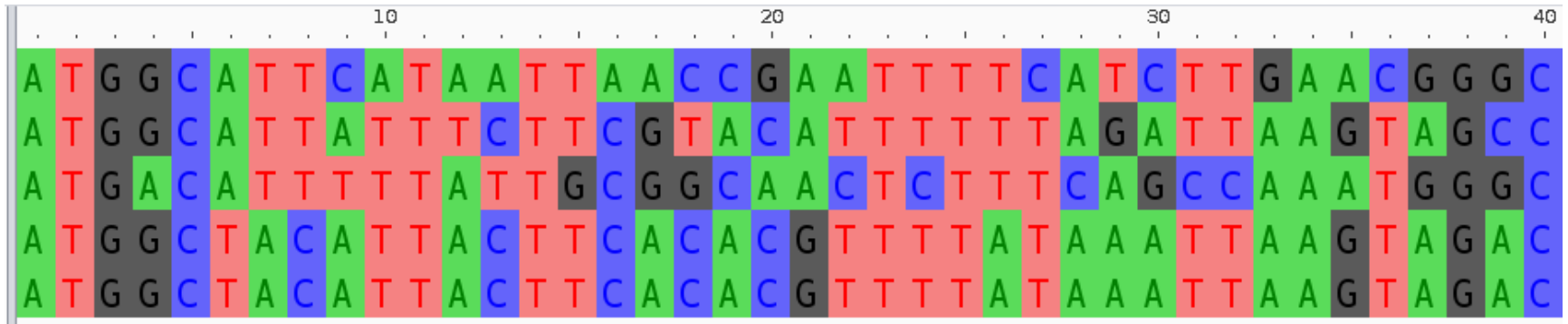


The branching points generate a tree structure

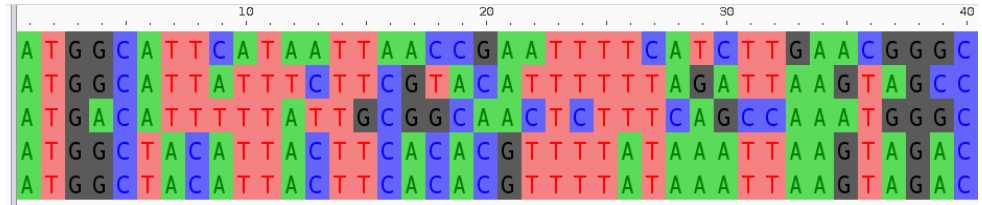
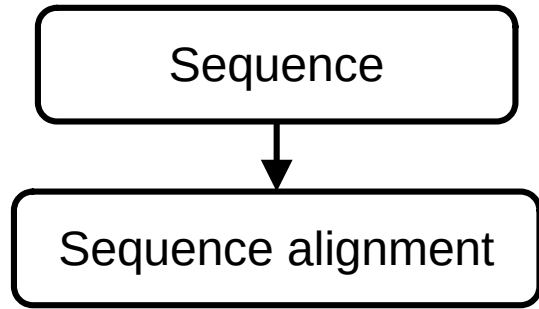
Phylogenetics using molecular data

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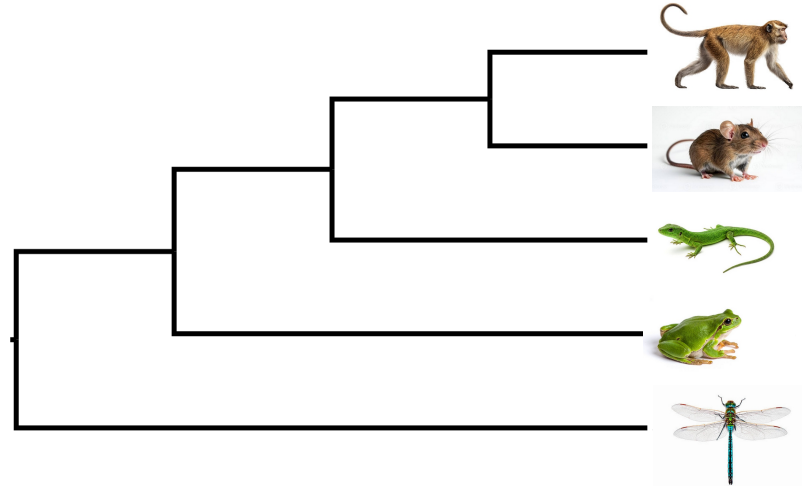
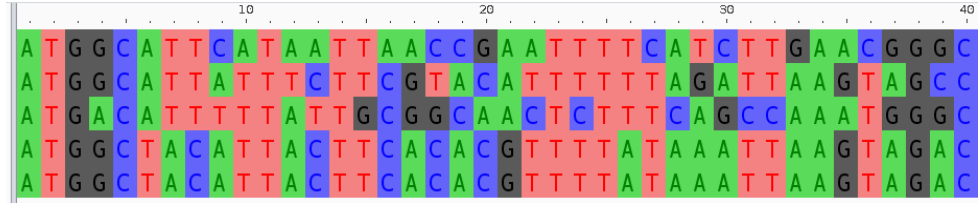
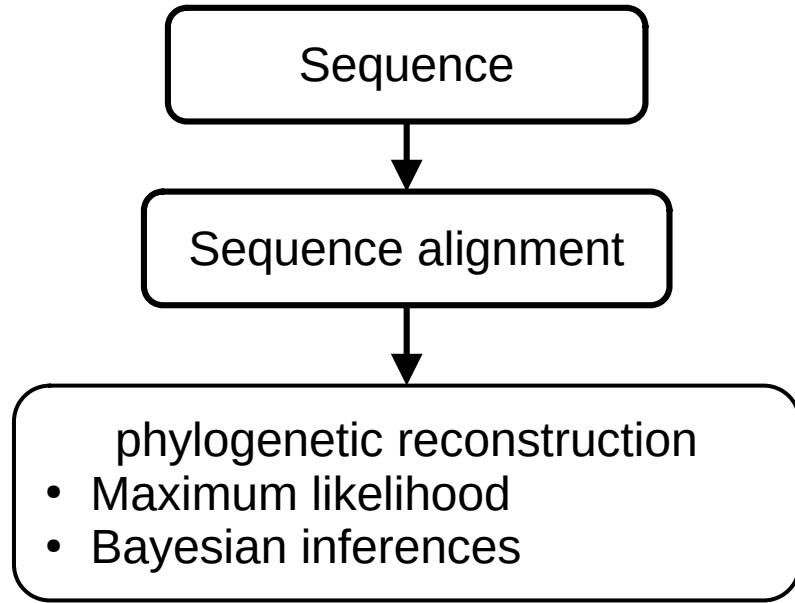
- DNA is replicated and vertically transmitted over the generations.
- Sequences are often highly specific
- Discrete variations
- Objectivity
- Abundance
- Ubiquity



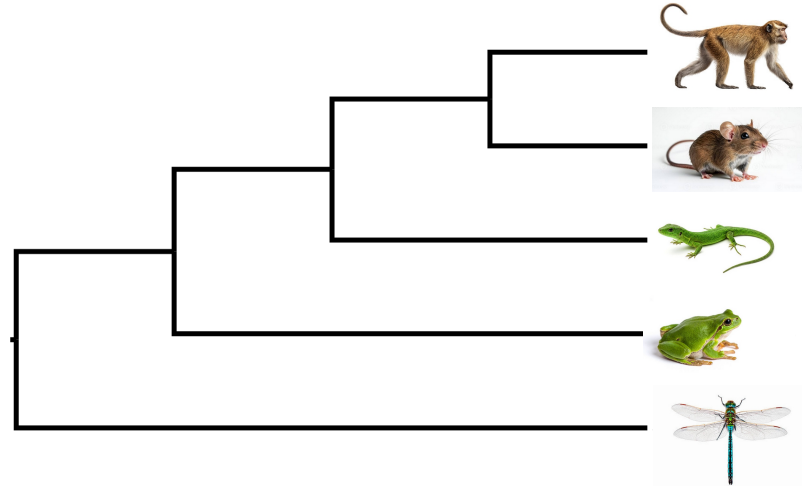
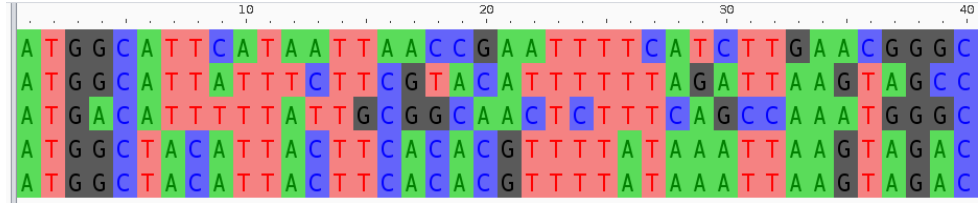
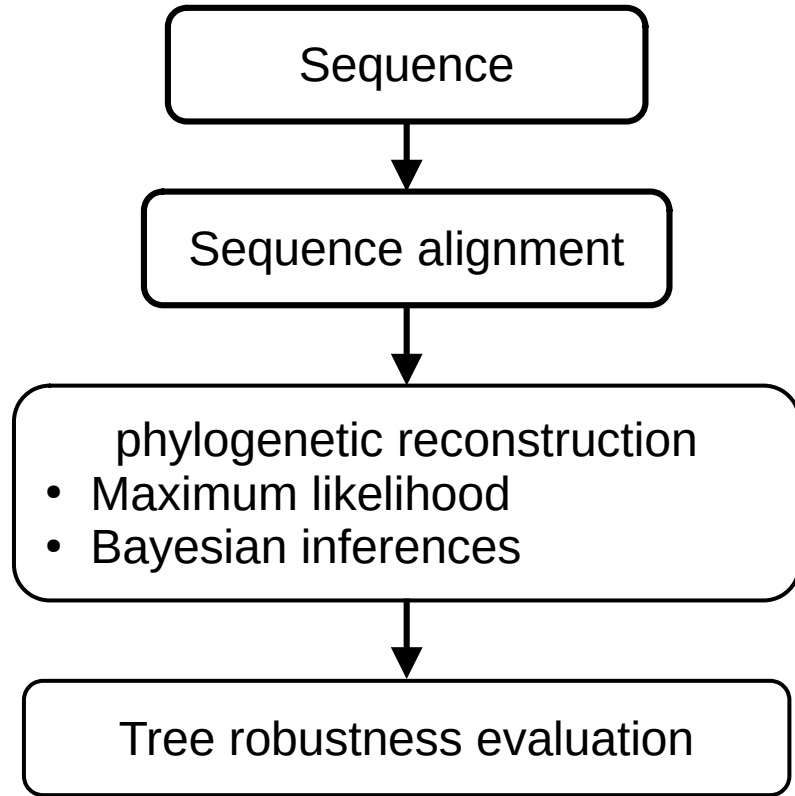
Phylogenetic reconstruction steps



Phylogenetic reconstruction steps



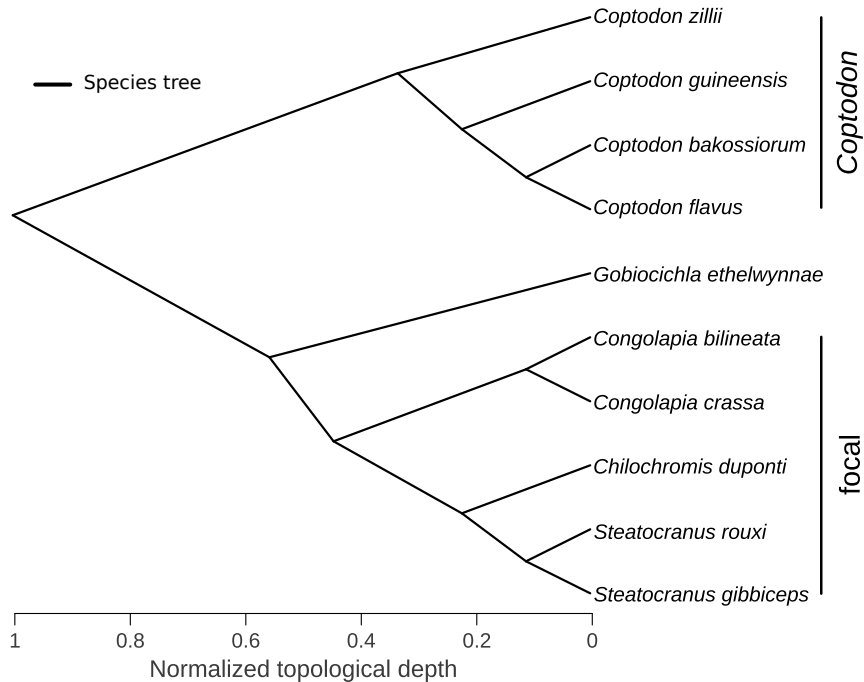
Phylogenetic reconstruction steps



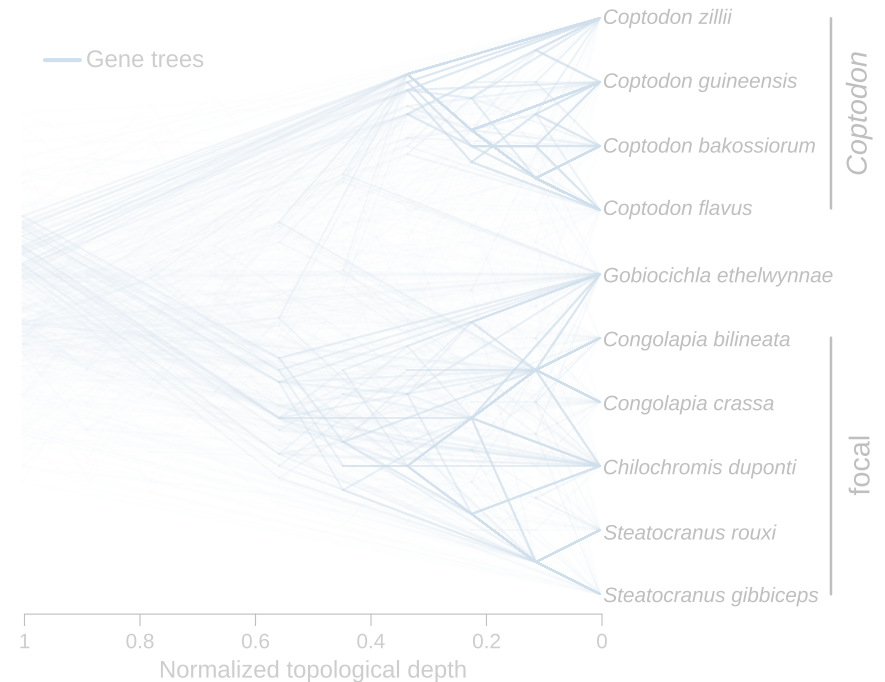
Molecular characters

With the improvement of sequencing technologies, hundreds or thousands of loci can be sequenced.

All the loci can be concatenated to produce a unique tree.



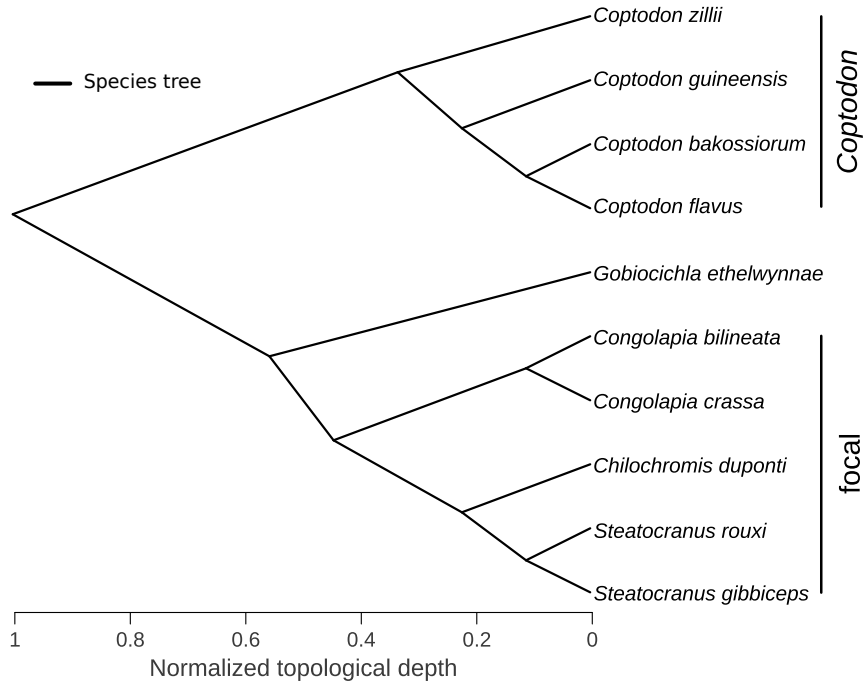
All the loci can be used independently to produce a tree per gene.



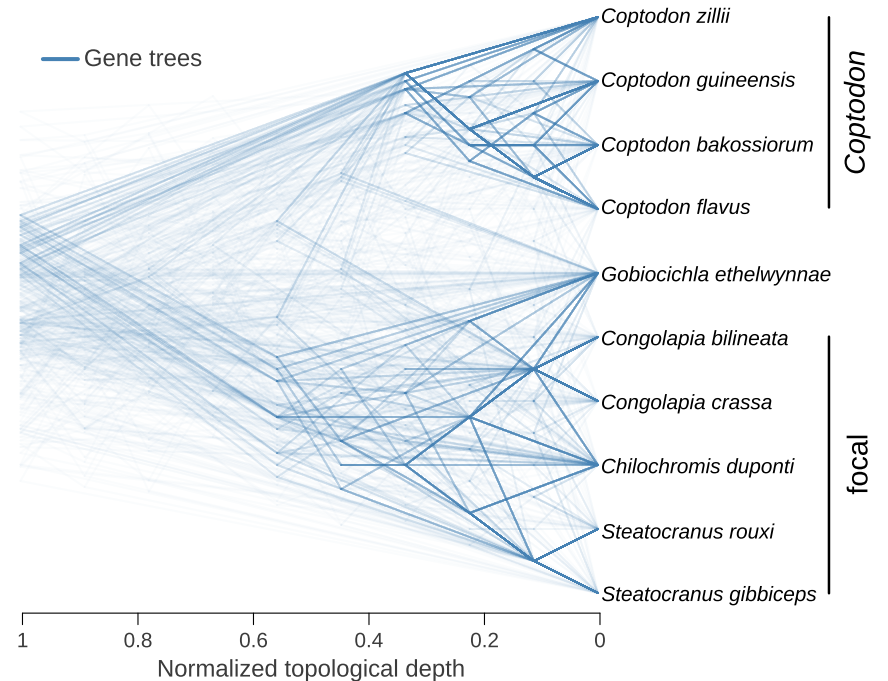
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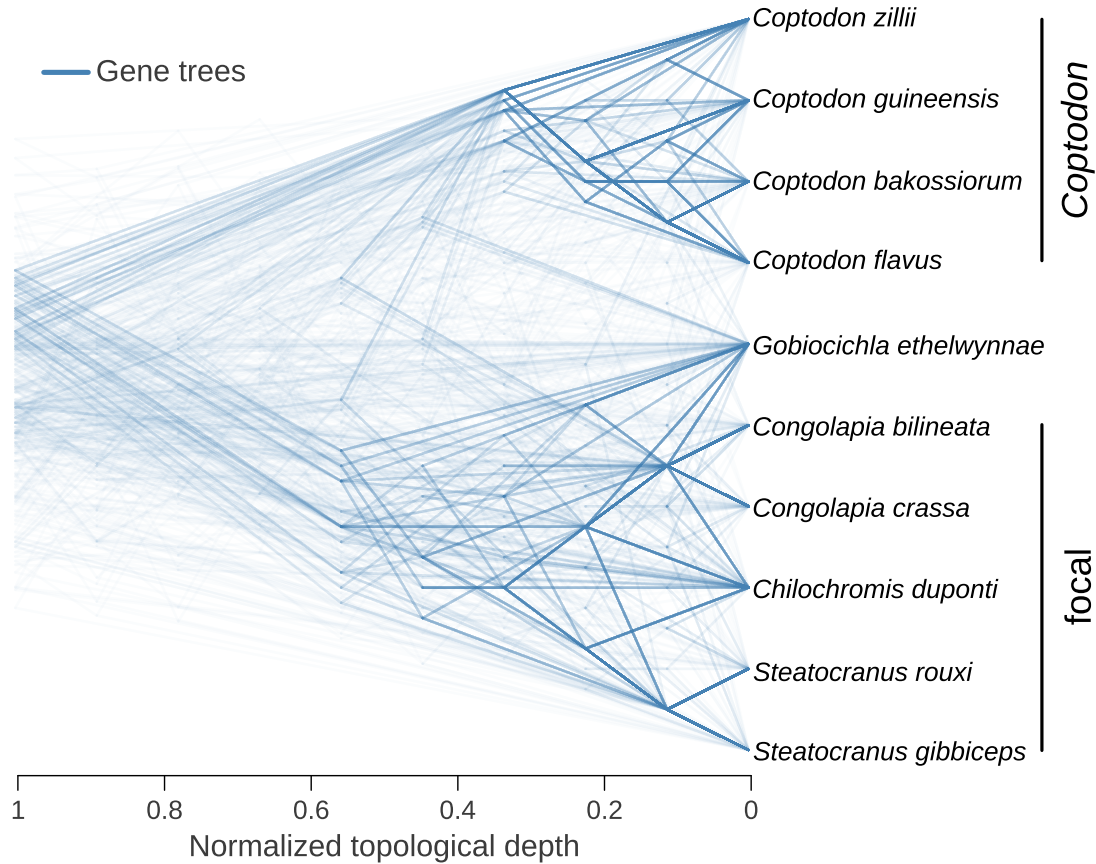
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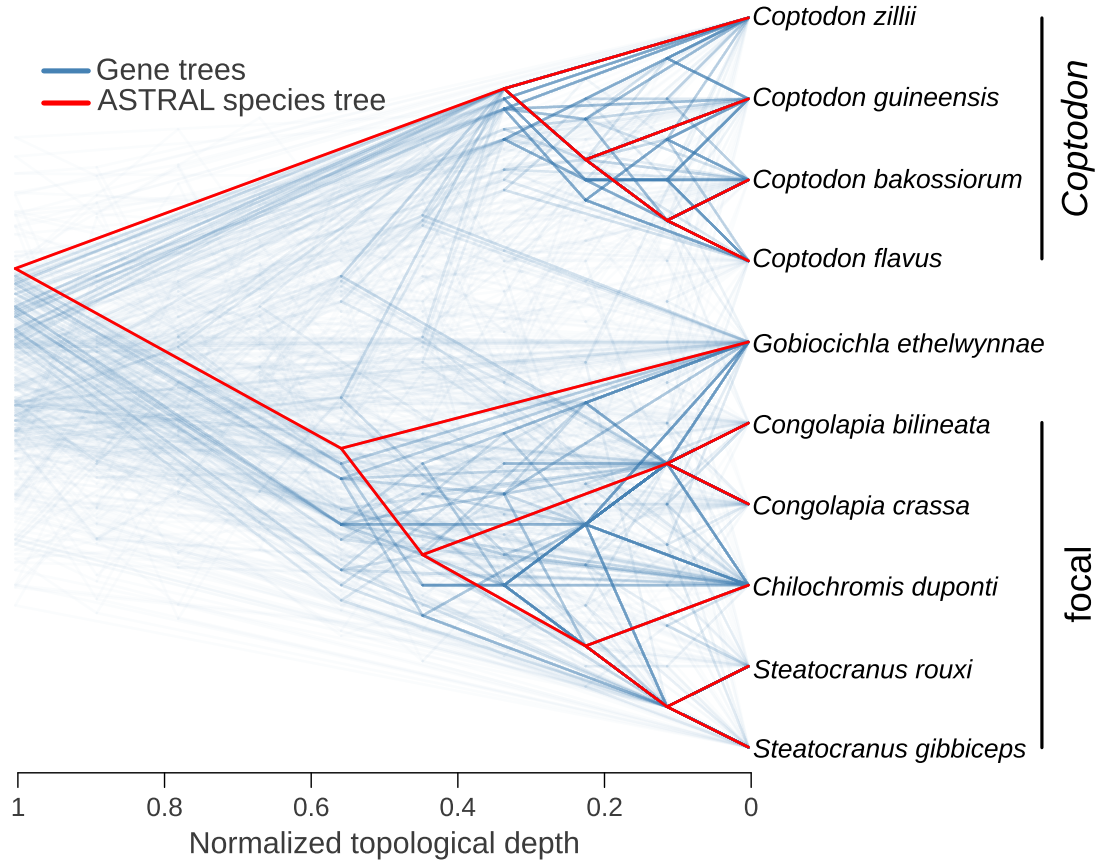


Species tree and gene trees



Gene tree represent the evolutionary history of a gene among species

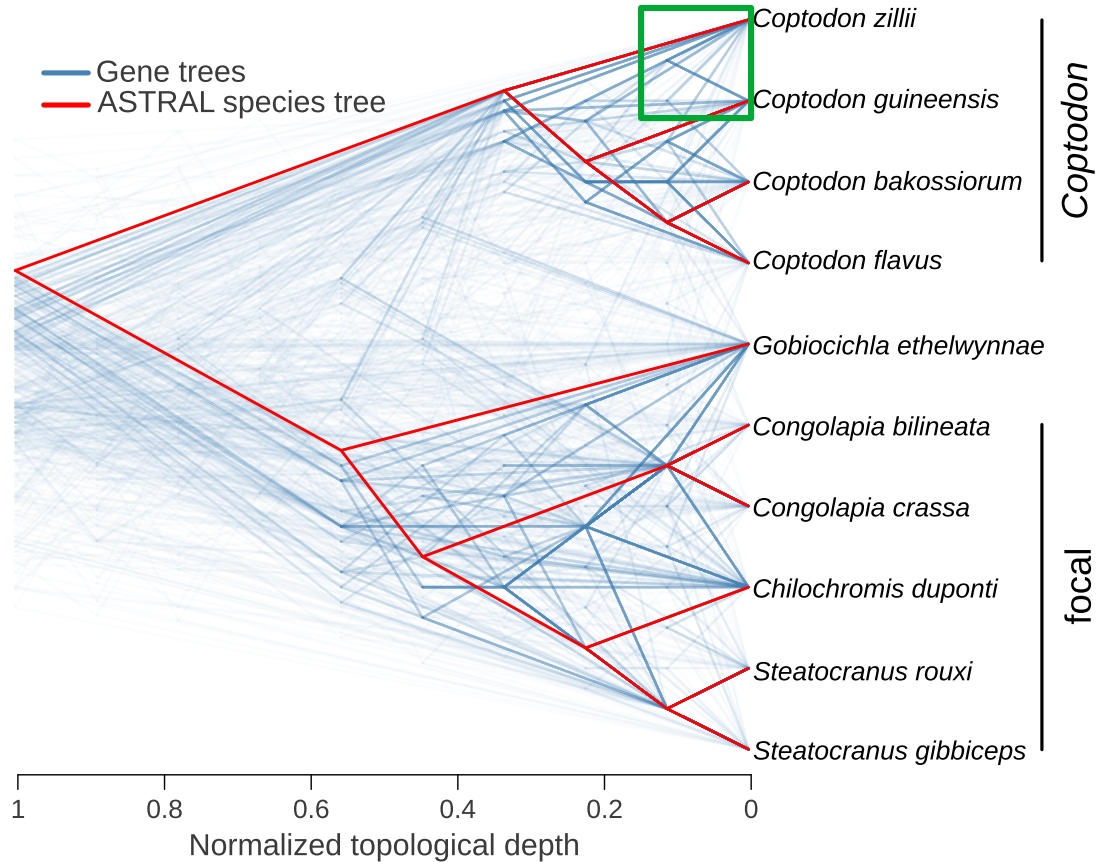
Species tree and gene trees



Gene tree represent the evolutionnary history of a gene among species

Species tree represent the averall evolutionary history of the species.

Species tree and gene trees



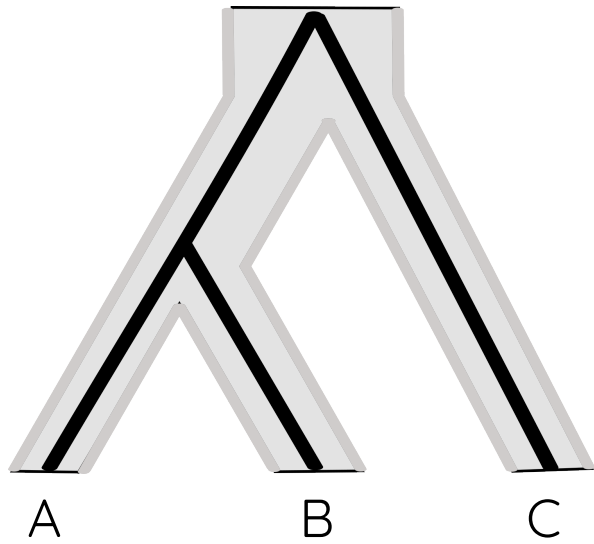
Gene trees represent the evolutionary history of a gene among species

The species tree represents the overall evolutionary history of the species.

Gene trees and species trees can differ in the connections between species and in the depth of the nodes.

Species tree and gene trees

One species tree but 3 possible gene trees

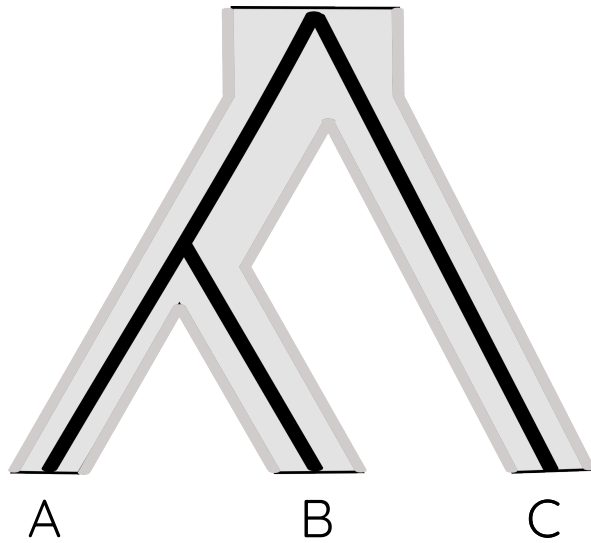


Concordant gene tree
((AB)C)

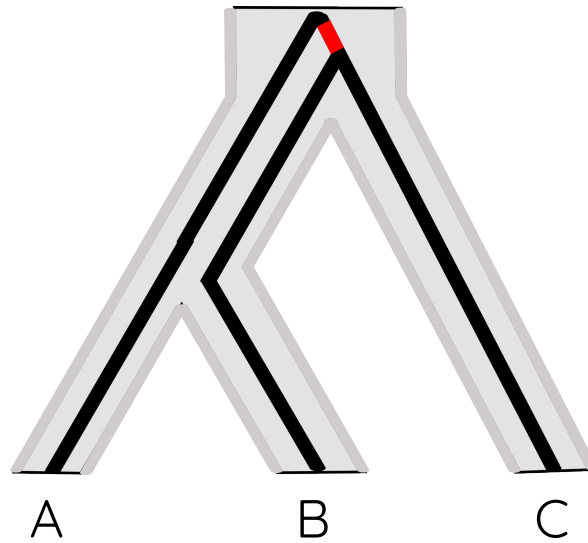


Species tree and gene trees

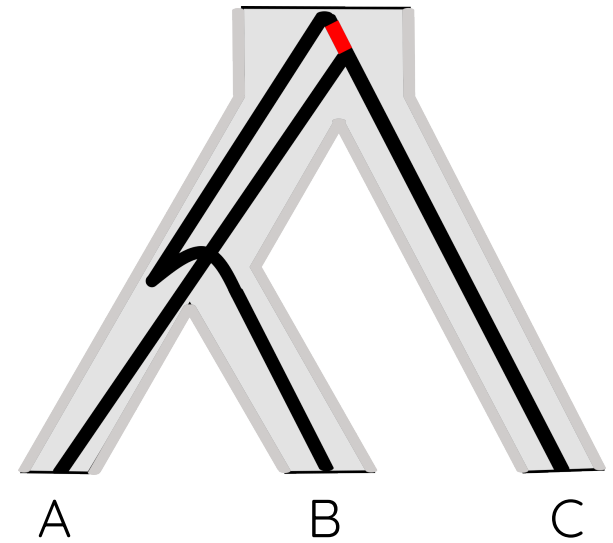
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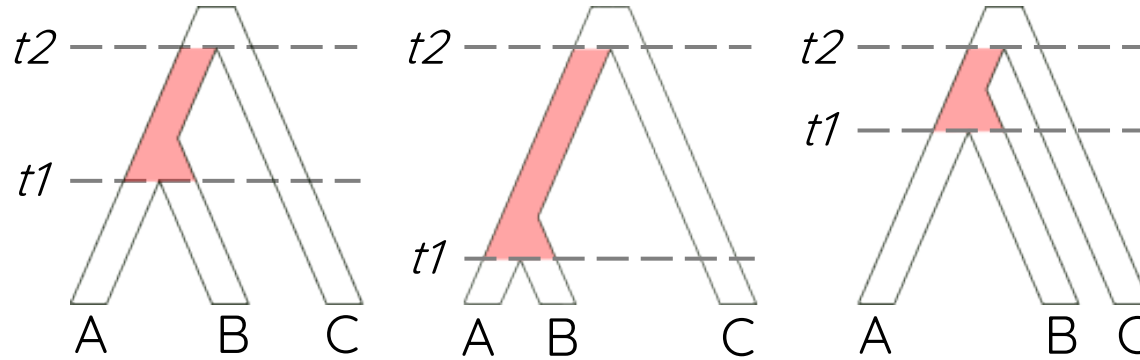


Discordant gene trees
 $((BC)A)$



$((AC)B)$

Incomplete lineage sorting



If multiple speciation events occur in a short period of time ($t_2 - t_1$ around N_e generations), polymorphism can be shared between the new species.

Incomplete lineage sorting

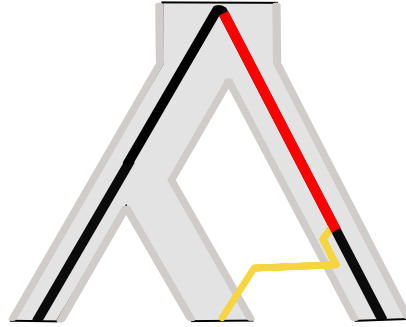


If multiple speciation events occur in a short period of time ($t_2 - t_1$ around N_e generations), polymorphism can be shared between the new species.

ILS properties :

- The fixed allele in each species is randomly chosen.
- Over all the genes affected by ILS, each topology can be observed in the same proportions.
- The total length of a tree tends to be longer than in a case without incomplete lineage sorting.

Gene flow

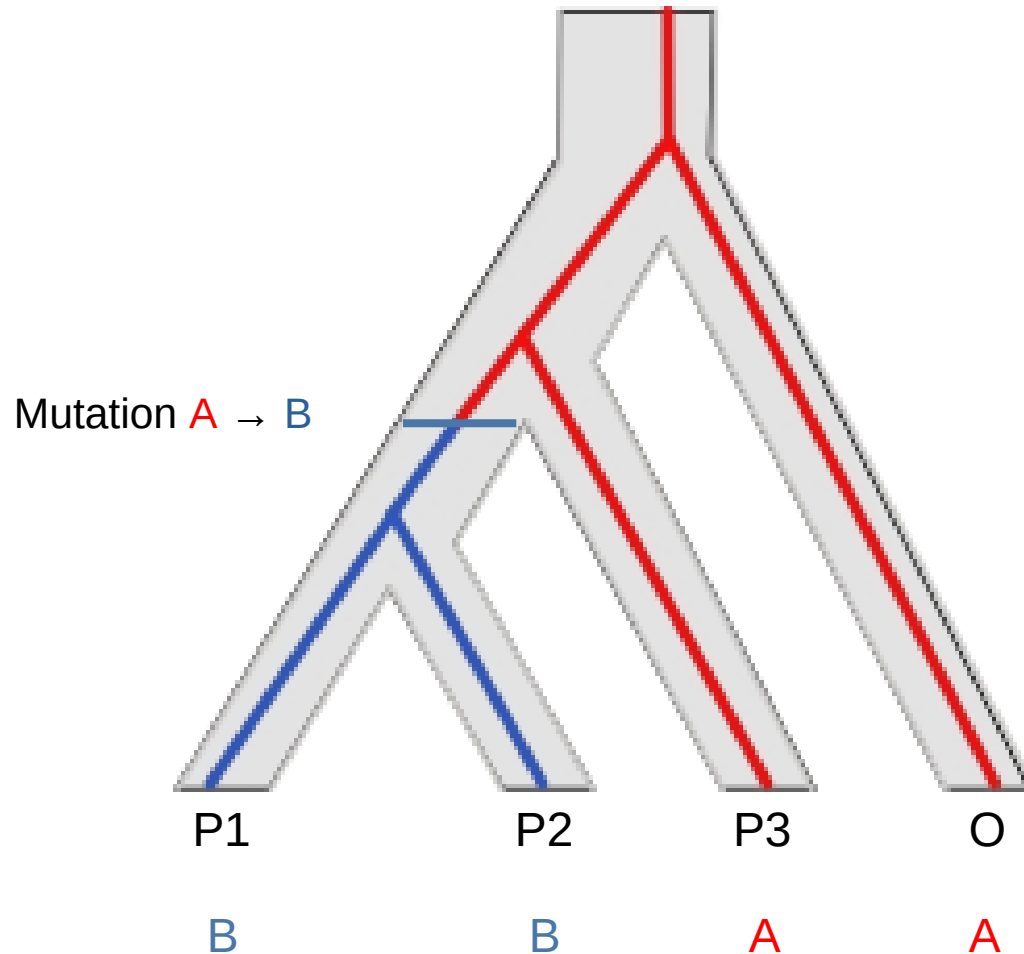


Gene flow between distinct species occurs whenever genetic material becomes integrated into a recipient species' gene pool.

Hybrids → Backcross → Introgression

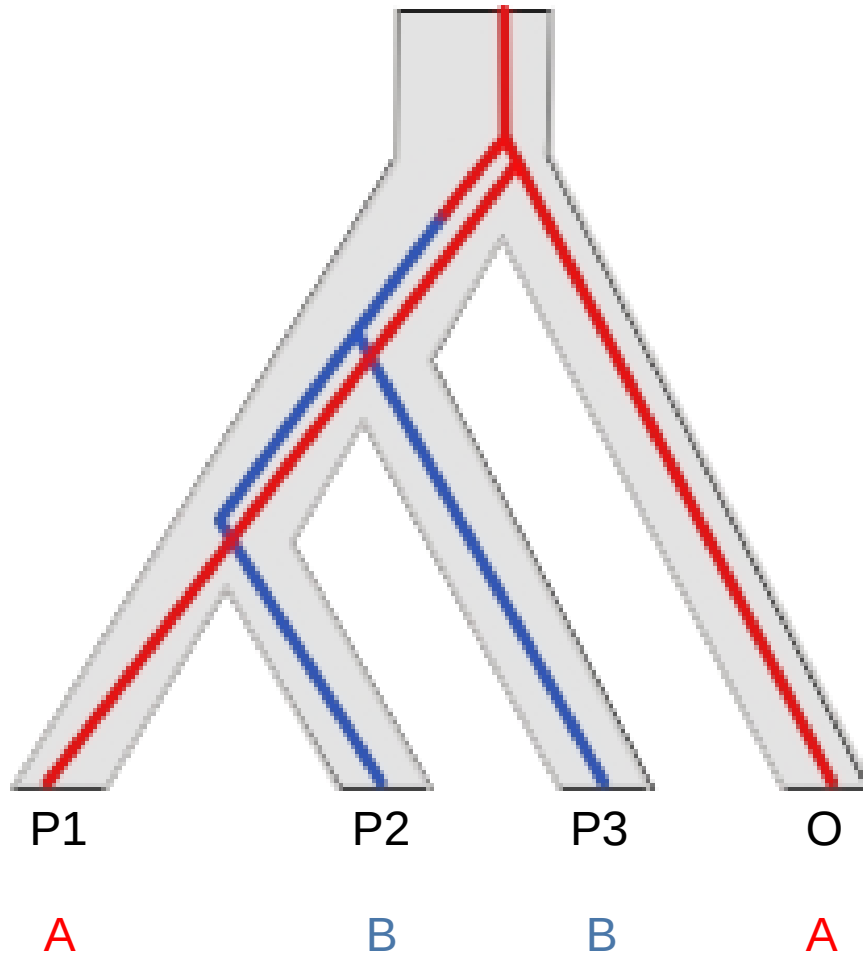
Unlike ILS, GF reduces the overall size of a phylogenetic tree.

Distinguish the GF from ILS using expected proportions (ABBA-BABA)



Concordant scenario $BBAA$.

Distinguish the GF from ILS using expected proportions (ABBA-BABA)

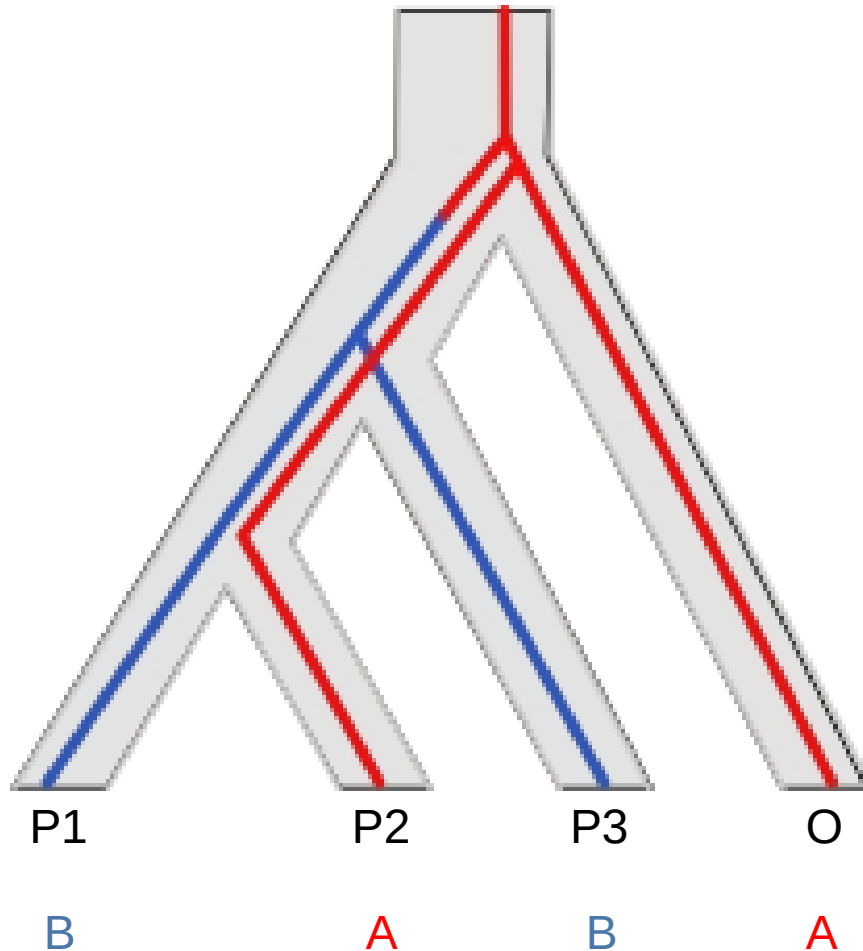


Concordant scenario BBAA.

ILS scenario, two discordant possibilities:

- ABBA

Distinguish the GF from ILS using expected proportions (ABBA-BABA)



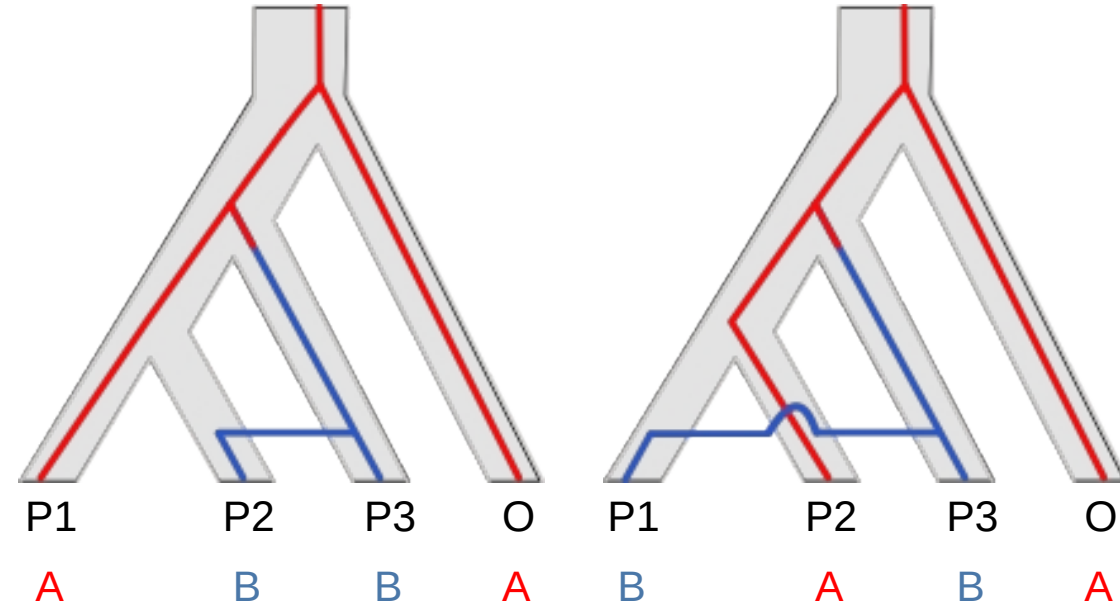
Concordant scenario BBAA.

ILS scenario, two discordant possibilities:

- ABBA
- BABA

Proportion ABBA $\approx$ proportion BABA

Distinguish the GF from ILS using expected proportions (ABBA-BABA)



Concordant scenario **BBAA**.

ILS scenario, two discordant possibilities:

- **ABBA**
- **BABA**

Proportion **ABBA** $\approx$ proportion **BABA**

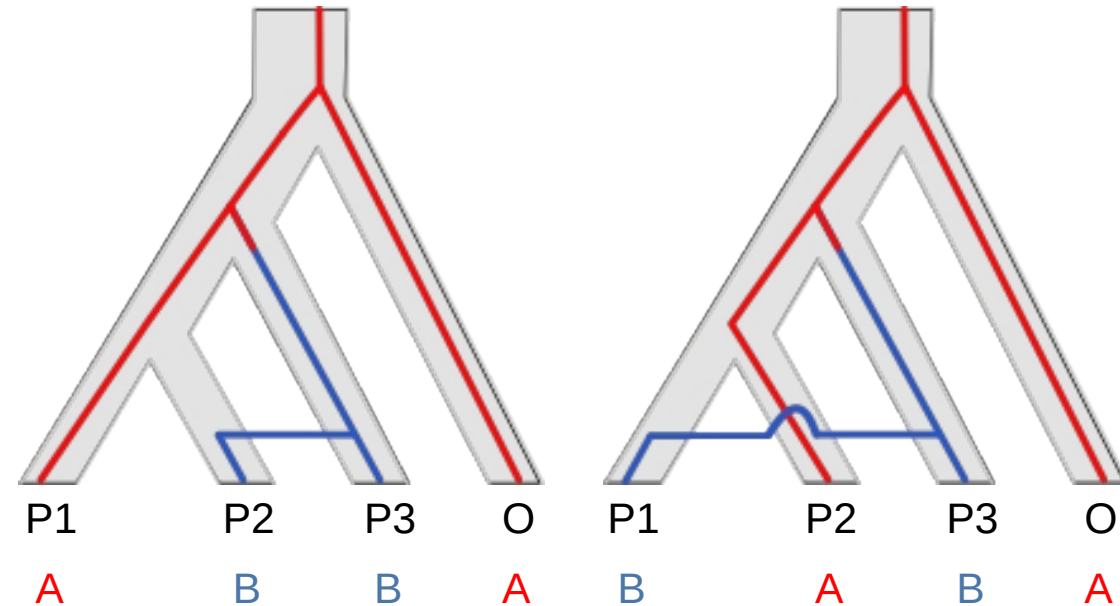
GF scenario, two discordant possibilities:

- **ABBA**
- **BABA**

Proportion **ABBA** $\gg$ proportion **BABA** or

Proportion **ABBA** $\ll$ proportion **BABA**

Distinguish the GF from ILS using expected proportions (ABBA-BABA)



$$D = \frac{n_{ABBA} - n_{BABA}}{n_{ABBA} + n_{BABA}}$$

Concordant scenario **BBAA**.

ILS scenario, two discordant possibilities:

- **ABBA**
- **BABA**

Proportion **ABBA** $\approx$ proportion **BABA**

GF scenario, two discordant possibilities:

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Distinguish the GF from ILS using expected proportions (ABBA-BABA)

ABBA-BABA test **limitations**:

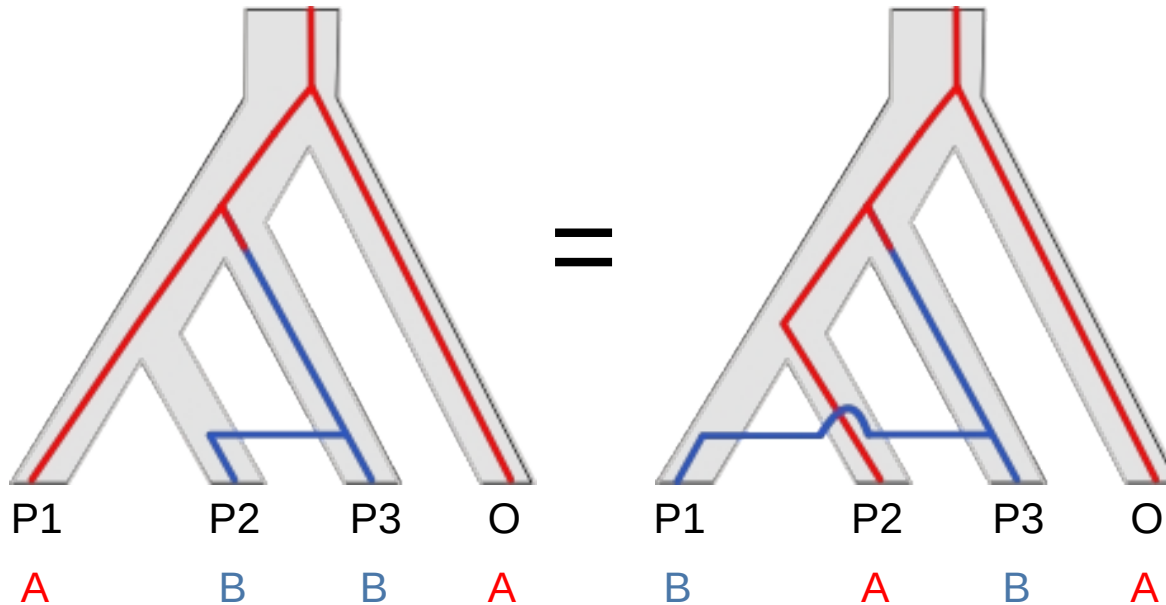
- Cannot date GF, only detect it

Distinguish the GF from ILS using expected proportions (ABBA-BABA)

ABBA-BABA test **limitations**:

- Cannot date GF, only detect it

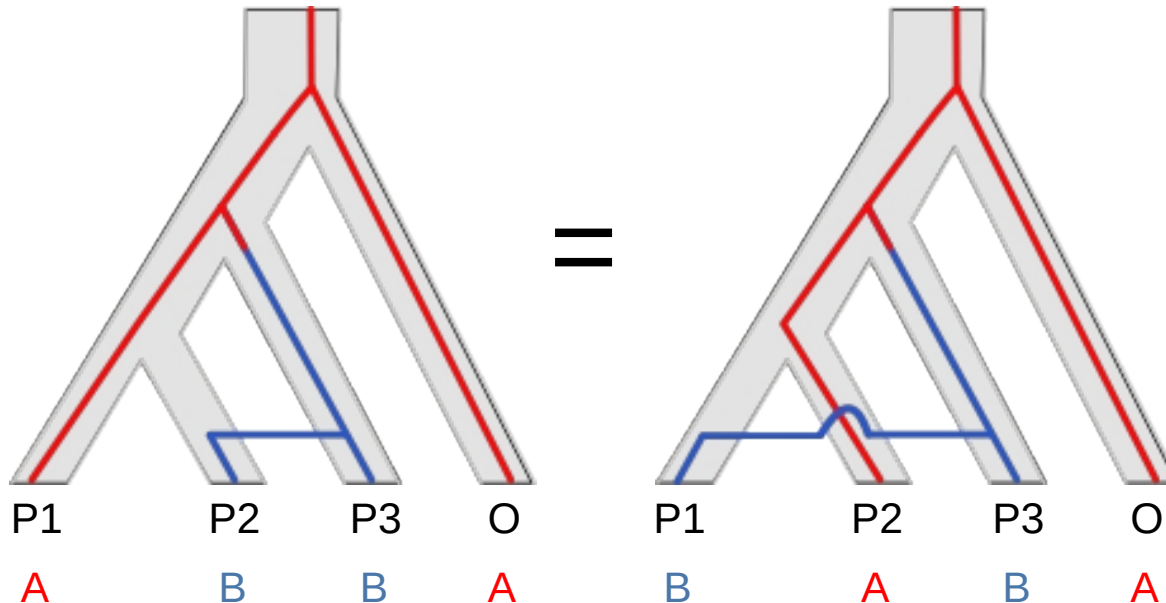
What happens if GF occurs between P3 and both P1 and P2?



Distinguish the GF from ILS using expected proportions (ABBA-BABA)

ABBA-BABA test **limitations**:

- Cannot date GF, only detect it.
- Cannot differentiate symmetrical GF from the ILS (Proportion **ABBA** $\approx$ proportion **BABA**).
==> This leads to an underestimation of GF



Distinguish the GF from ILS using expected branch length (QUIBL)

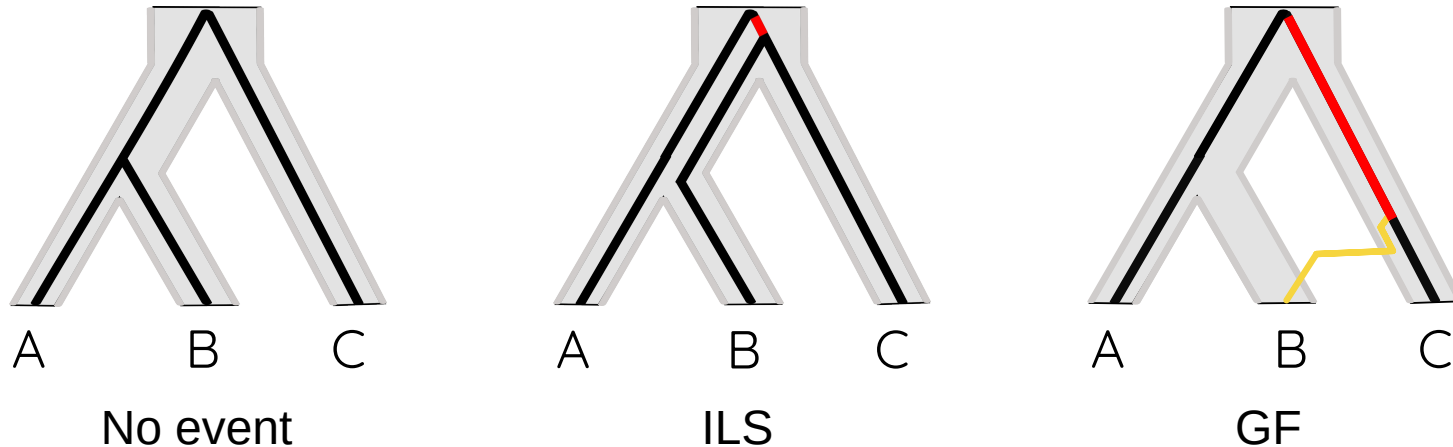
Genomic architecture and introgression shape a butterfly radiation

NATHANIEL B. EDELMAN , PAUL B. FRANDSEN , MIRIAM MIYAGI , BERNARDO CLAVIJO , JOHN DAVEY , REBECCA B. DIKOW , GONZALO GARCÍA-ACCINELLI

, STEVEN M. VAN BELLEGHEM , NICK PATTERSON , [...], AND JAMES MALLET  +19 authors [Authors Info & Affiliations](#)

SCIENCE • 1 Nov 2019 • Vol 366, Issue 6465 • pp. 594-599 • DOI: 10.1126/science.aaw2090

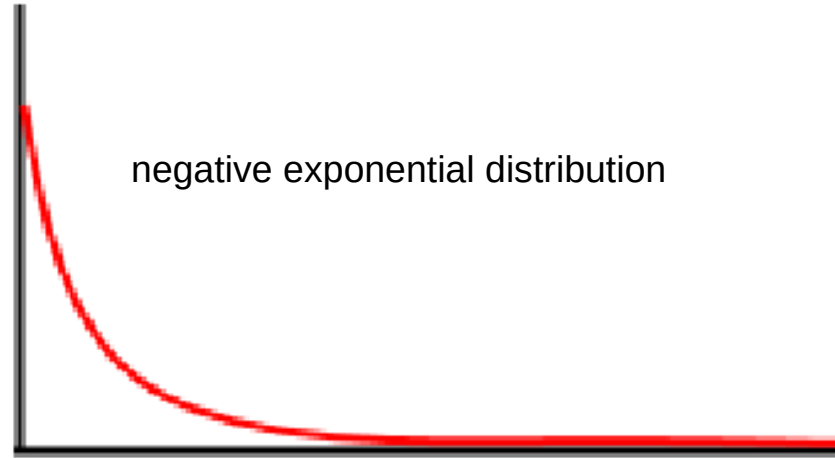
QUIBL: Quantifying Introgression via Branch Lengths



Distinguish the GF from ILS using branch length distribution (QUIBL)

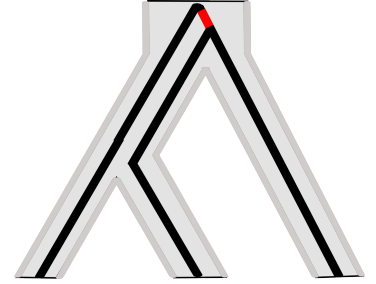
QuiBL: **Q**uantifying **I**ntrogression via **B**ranched **L**engths

Infer the distribution of the internal branch length



internal branch length

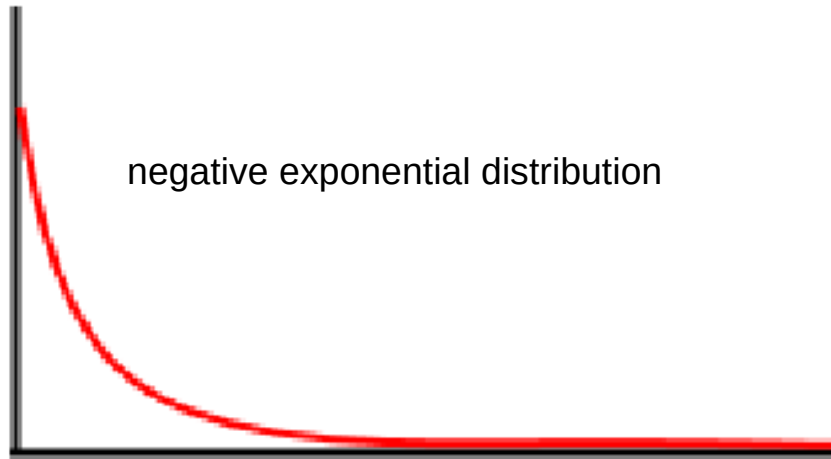
ILS



Distinguish the GF from ILS using branch length distribution (QUIBL)

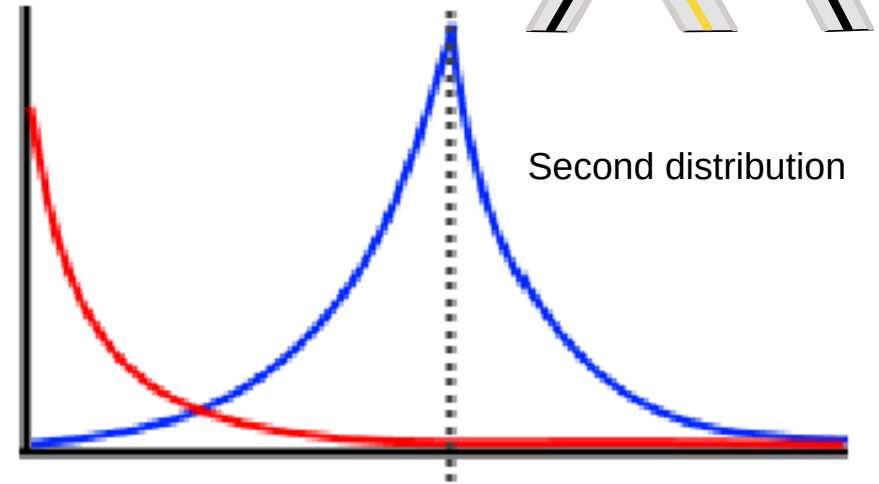
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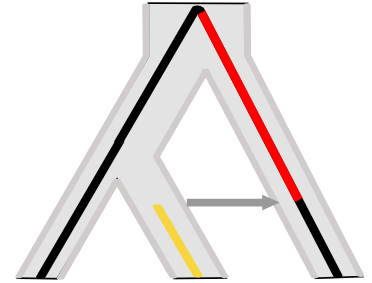
internal branch length

ILS



internal branch length

ILS + GF



Distinguish the GF from ILS using branch length distribution (QUIBL)

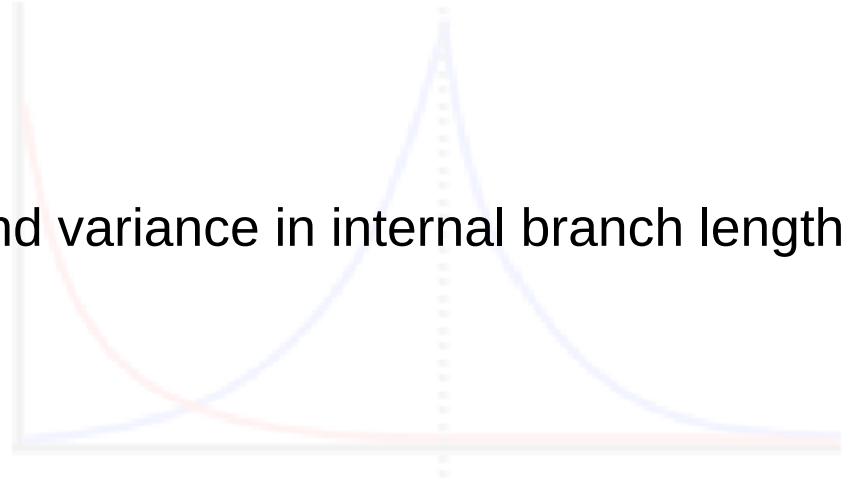
QUIBL: **Q**uantifying **I**ntrogression via **B**ranched **L**engths

Infer the distribution of the internal branch length

QUIBL **detects** an increased mean and variance in internal branch length



ILS



ILS + GF

Distinguish the GF from ILS using branch length distribution (QUIBL)

QUIBL: Quantifying Introgression via Branch Lengths

QUIBL **assumes**:

- Neutral loci
- Non-recombinant loci
- Constant population size
- Constant mutation rate among loci

QUIBL **benefits**:

- Detect symmetrical and asymmetrical GF
- Quantify GF

Distinguish the GF from ILS using branch length distribution (QUIBL)

QUIBL: Quantifying Introgression via Branch Lengths

QUIBL limitations:

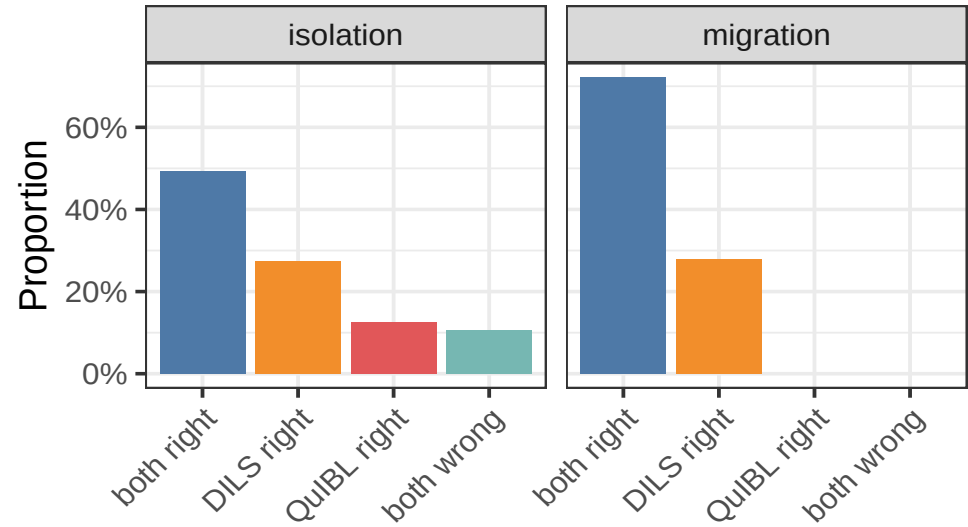
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QuiBL **limitations**:

- Constant mutation rate among loci
- Only uses internal branch
- Highly conservative

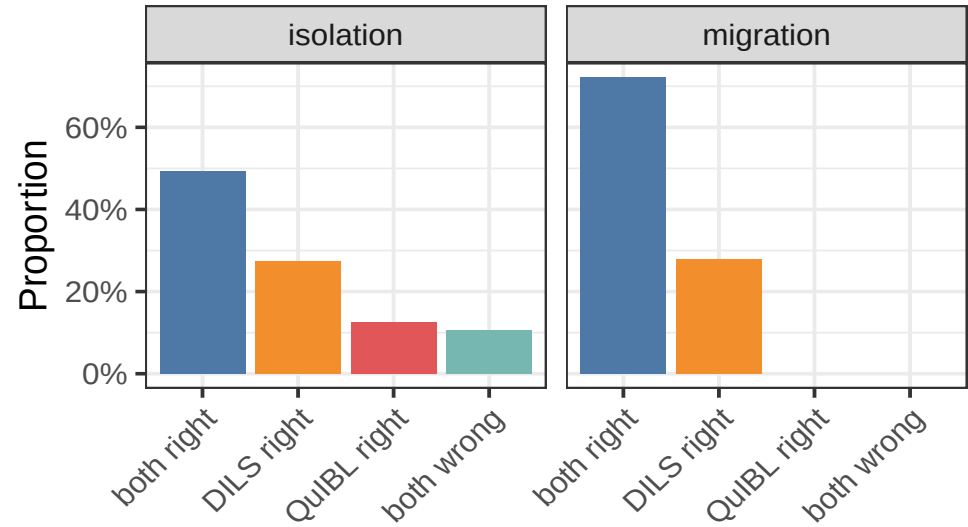


Distinguish the GF from ILS using branch length distribution (QUIBL)

QuiBL: **Q**uantifying **I**ntrogression via **B**ranched **L**engths

QuiBL **limitations**:

- Constant mutation rate among loci
- Only uses internal branch
- Highly conservative
- Gene flow between sister species cannot be inferred



Distinguish the GF from ILS using coalescence theory (Aphid)

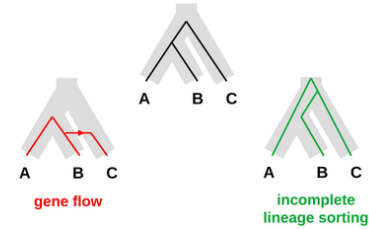
Topic: [Biophysics and computational biology](#), [Evolution](#), [Genetics/genomics](#)

An approximate likelihood method reveals ancient gene flow between human, chimpanzee and gorilla

Galtier, Nicolas¹  

Corresponding author(s): Galtier, Nicolas (nicolas.galtier@umontpellier.fr)

10.24072/pcjournal.359 - Peer Community Journal, Volume 4 (2024), article no. e3



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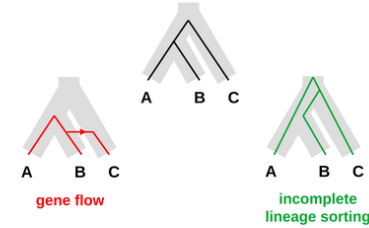
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Like QuIBL, **Aphid**:

- Works using a triplet of species and outgroups
- Uses branch length to distinguish and quantify GF and ILS
- Assumes:
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 - Non-recombinant loci
 - Constant population size

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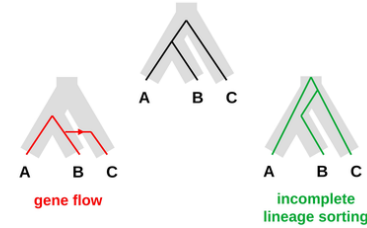
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- Assumes:
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 - Non-recombinant loci
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But **Aphid**:

- Corrects the mutation rate per locus
- Use internal and terminal branch lengths
- Compares these lengths to the expectation under the coalescence theory

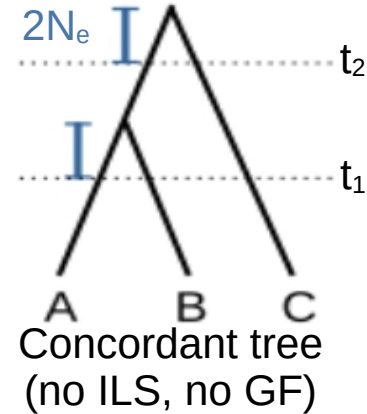
Scenarios tested by Aphid and coalescence times

A gene tree can fall into one of the three possible scenarios:

- No ILS and no GF (no event)

$$t_{A|B} = t_1 + 2N_e$$

$$t_{(A,B)|C} = t_2 + 2N_e$$



Scenarios tested by Aphid and coalescence times

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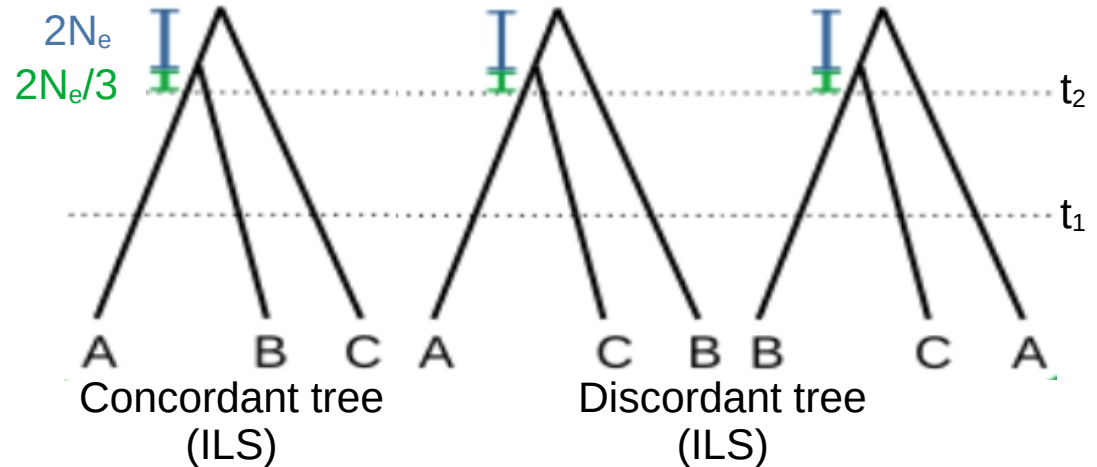
$$t_{A|B} = t_1 + 2N_e$$

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- Incomplete lineage sorting

$$t_{\text{first}} = t_2 + 2N_e/3$$

$$t_{\text{second}} = t_2 + 2N_e/3 + 2N_e$$



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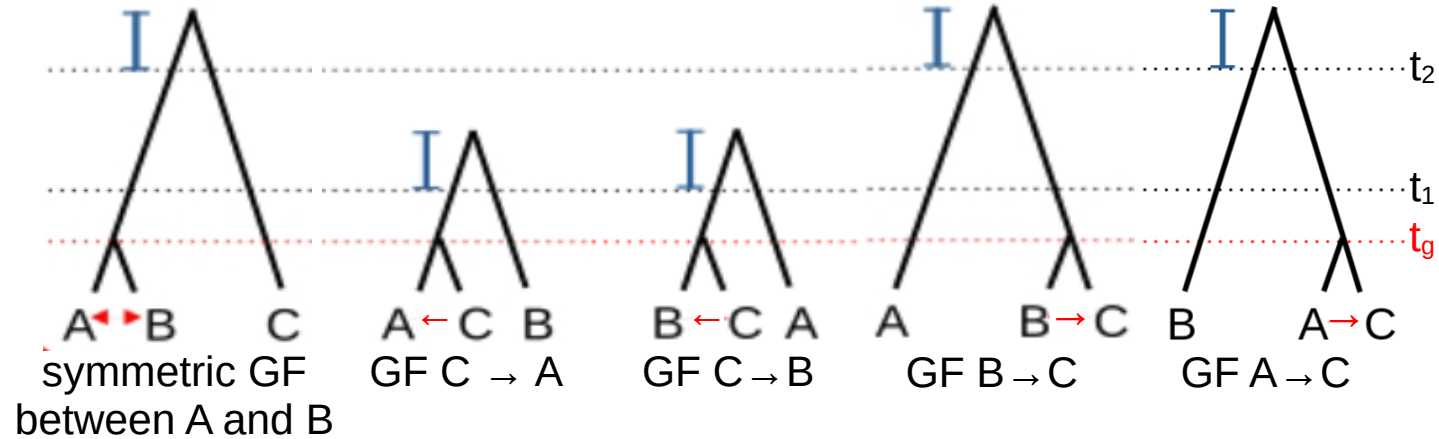
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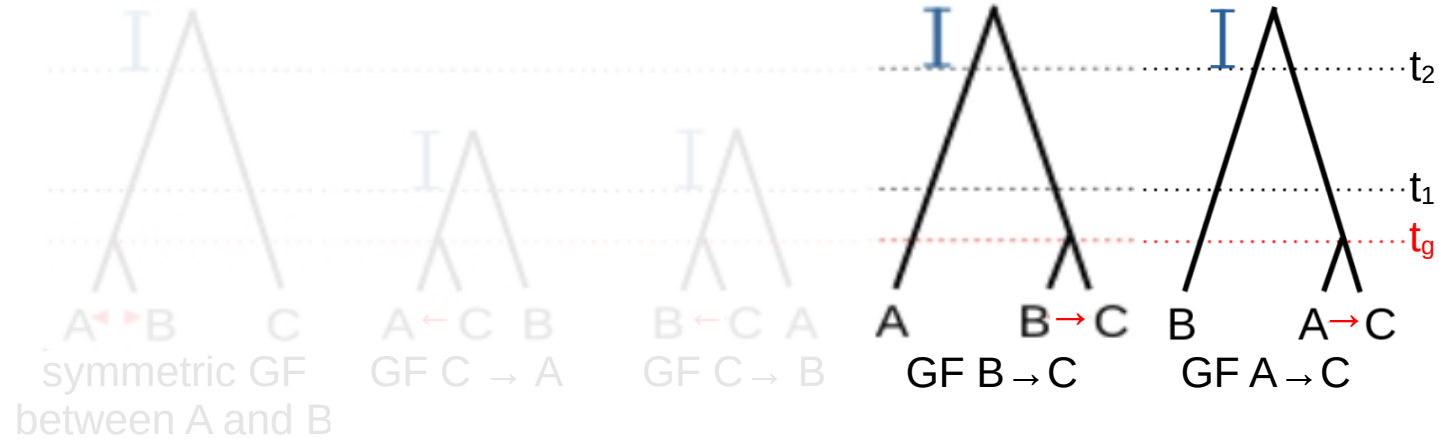
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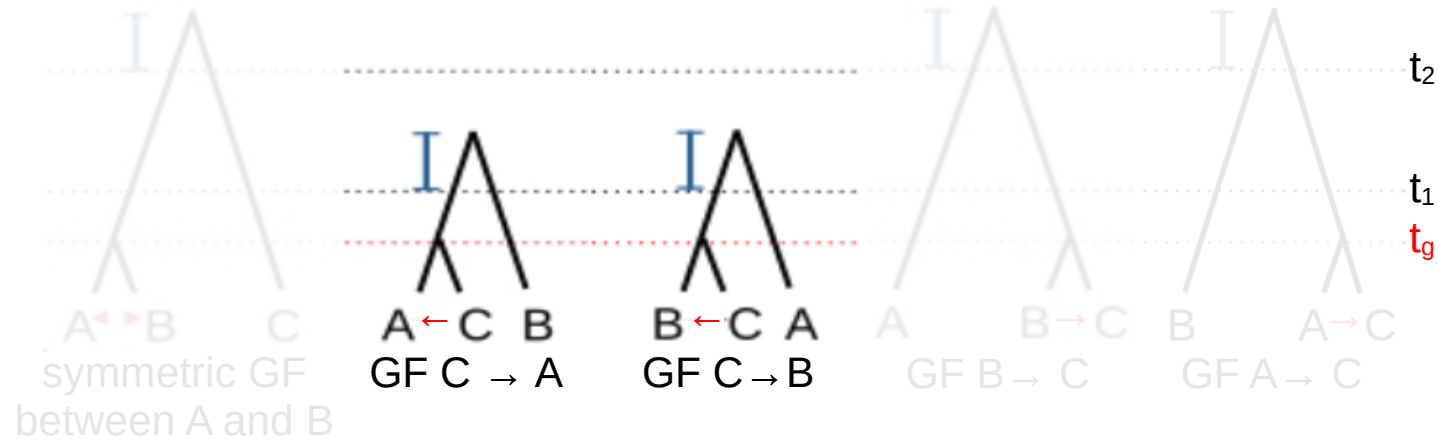
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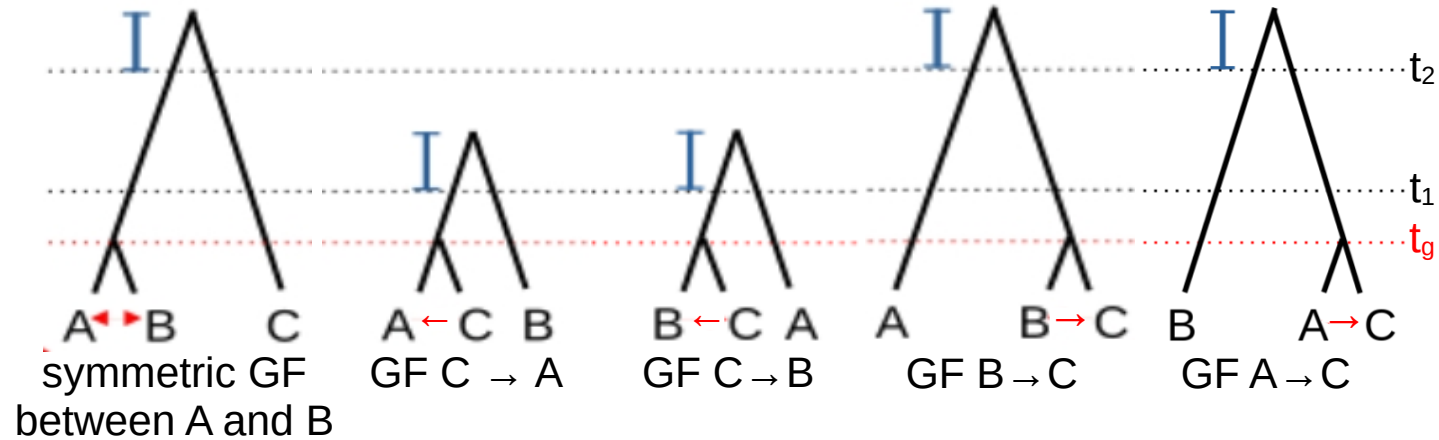
$$t_{\text{first}} = t_2 + 2N_e/3$$

$$t_{\text{second}} = t_2 + 2N_e/3 + 2N_e$$

- Gene flow

$$t_{\text{first}} = t_g$$

$$t_{\text{second}} = t_2 + 2N_e \text{ or } t_1 + 2N_e$$



Scenarios tested by Aphid and coalescence times

A gene tree can fall into one of the three possible scenarios:

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$$t_{A|B} = t_1 + 2N_e$$

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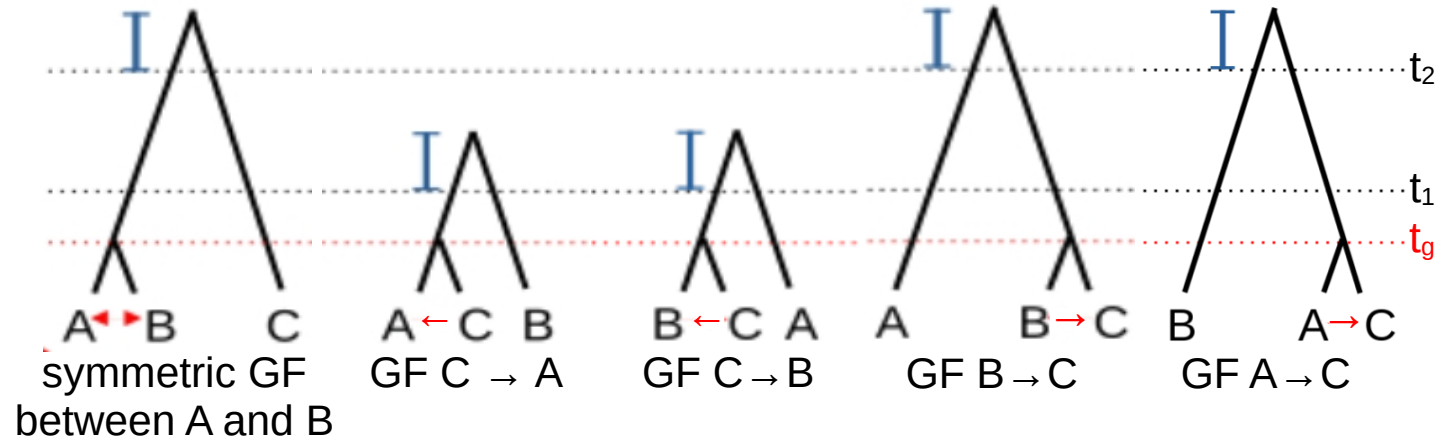
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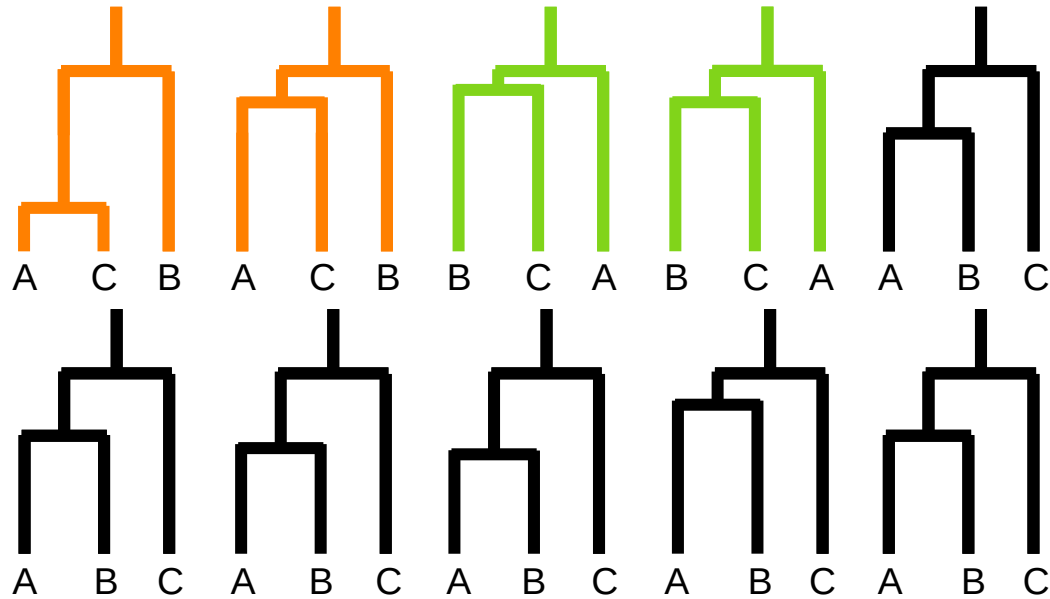
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How Aphid works

To perform its analysis, Aphid needs a set of gene trees associated with the sequence length for each gene.

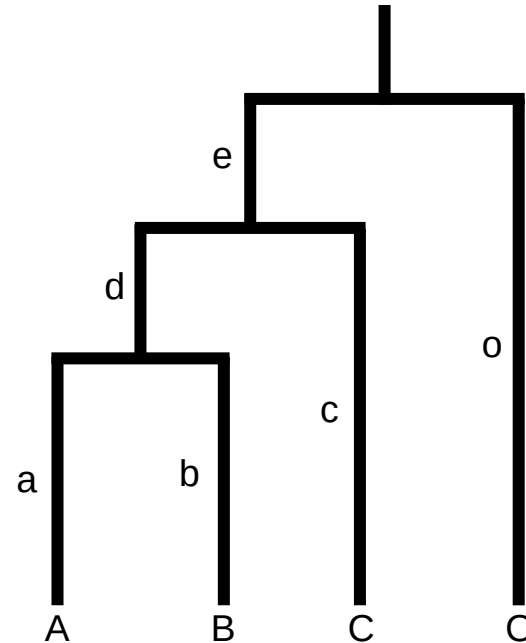


How Aphid works

- 1) Assign prior probabilities to each evolutionary scenario.

Calculate the mean length of the different paths from root to leaves for a gene g (l_g).

$$l_g = \frac{(a+d+e) + (b+d+e) + (c+e) + o}{4}$$



How Aphid works

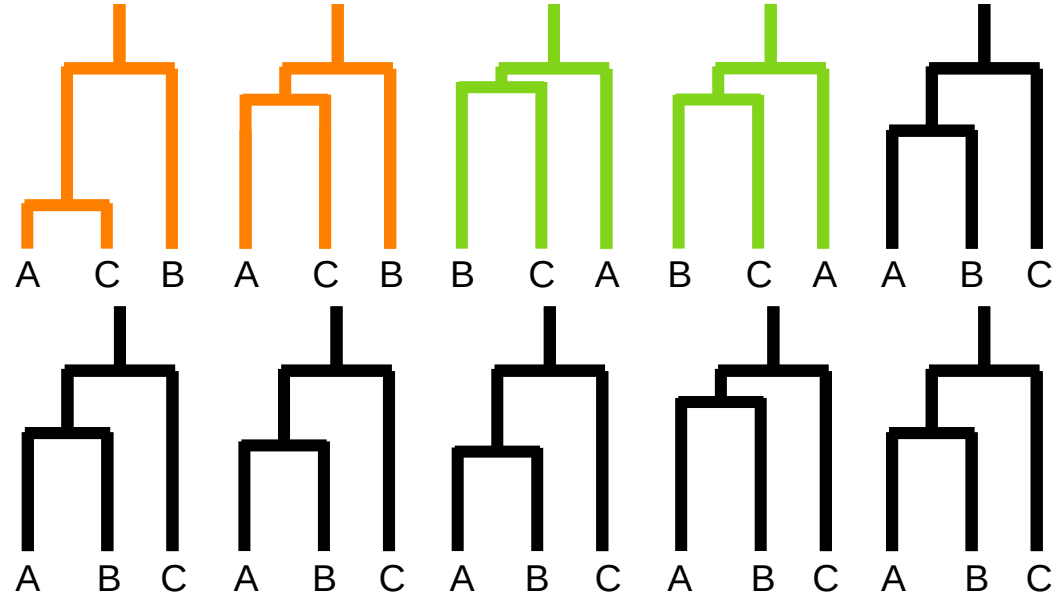
- 1) Assign prior probabilities to each evolutionary scenario.

The global mean length from root to leaves over the G genes is calculated as:

$$L = \frac{\sum l_g}{G}$$

The relative mutation rate can be calculated per gene:

$$\alpha_g = \frac{l_g}{L}$$



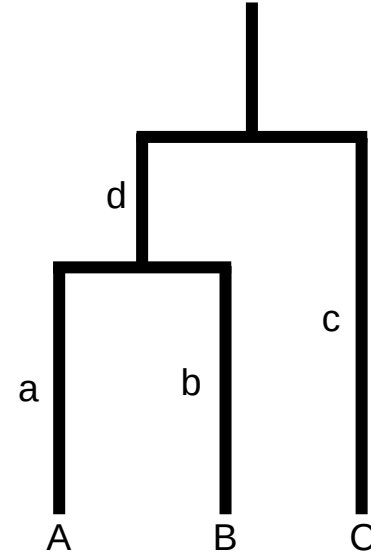
How Aphid works

- 1) Assign prior probabilities to each evolutionary scenario.

The number of mutations per branch is calculated as:

$$n_g(bl) = sl_g \times bl \quad \text{with } bl \in [a, b, c, d]$$

Sequence length



How Aphid works

1) Assign prior probabilities to each evolutionary scenario.

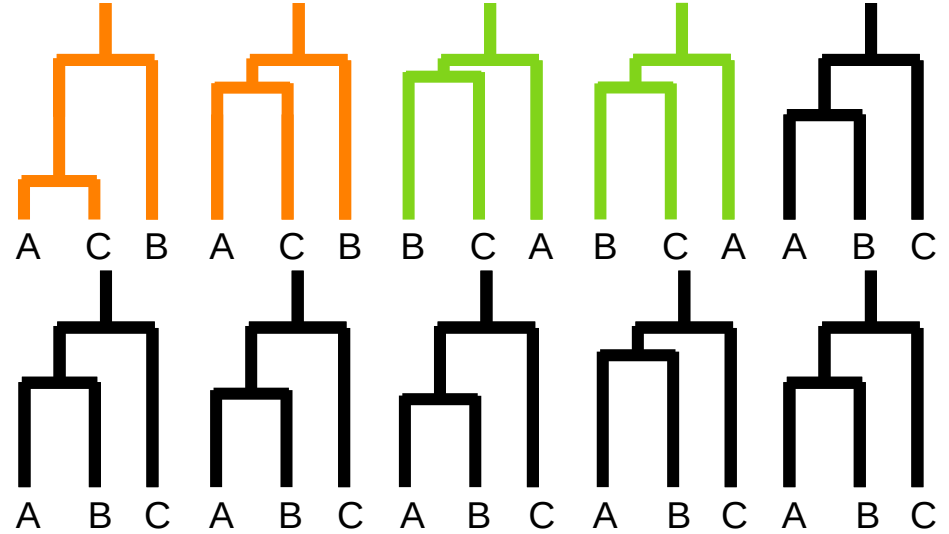
The number of mutations per branch is calculated as:

$$n_g(bl) = sl_g \times bl$$

Then calculate the mean number of mutations per branch among the genes:

$$m_a = \frac{1}{G} \sum_g n_g(a_g) \quad m_b = \frac{1}{G} \sum_g n_g(b_g)$$

$$m_c = \frac{1}{G} \sum_g n_g(c_g) \quad m_d = \frac{1}{G} \sum_g n_g(d_g)$$



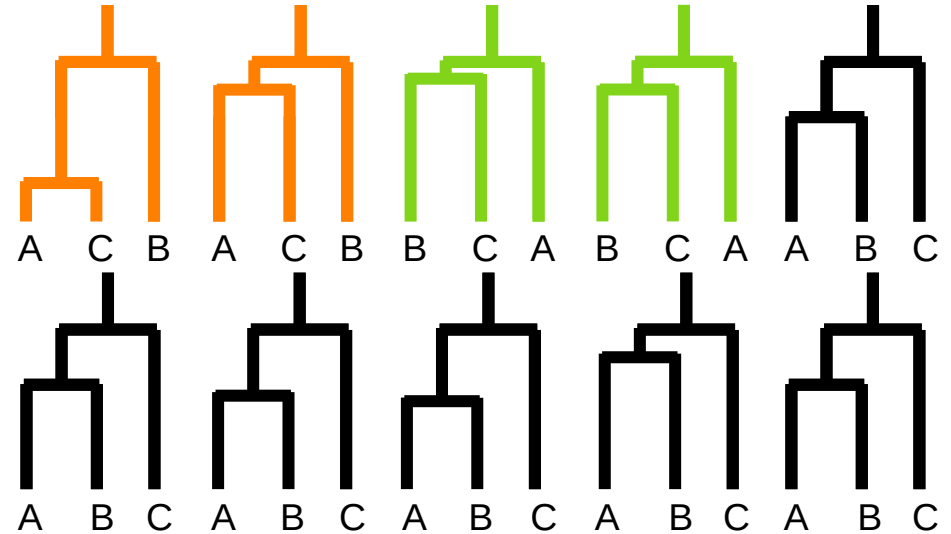
How Aphid works

1) Assign prior probabilities to each evolutionary scenario.

Calculate the estimated speciation times:

$$\tau_1 \frac{m_a + m_b - \theta}{2}$$

$$\tau_2 \frac{m_c + m_d + \tau_1 - \theta}{2}$$



How Aphid works

1) Assign prior probabilities to each evolutionary scenario.

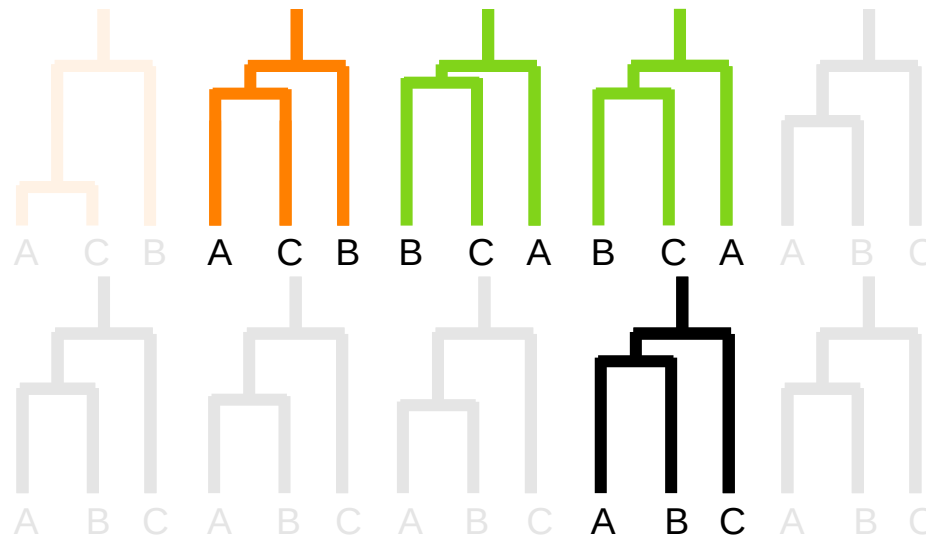
Calculate the estimated speciation times:

$$\tau_1 \frac{m_a + m_b - \theta}{2}$$

$$\tau_2 \frac{m_c + m_d + \tau_1 - \theta}{2}$$

Then the prior probability of observing ILS among the trees:

$$p_{ILS} = e^{2 \frac{\tau_1 - \tau_2}{\theta}}$$



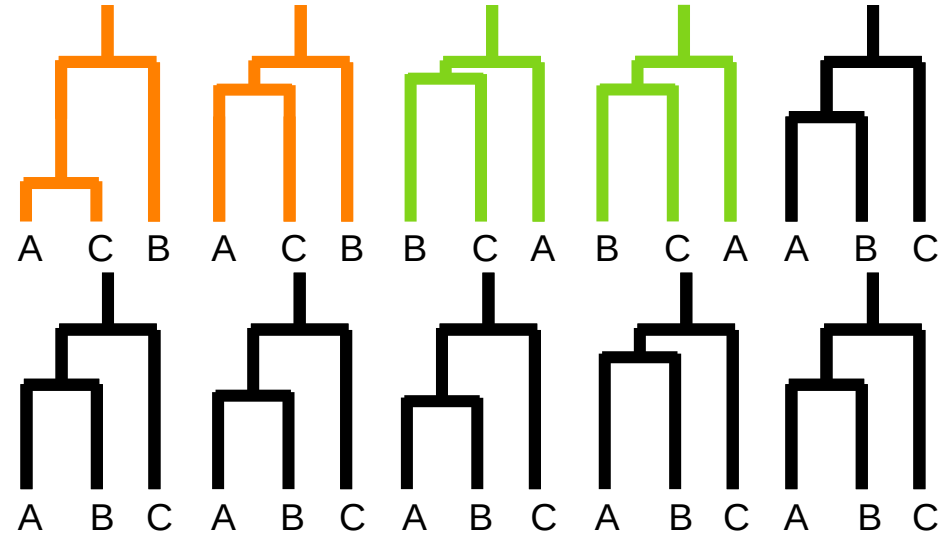
How Aphid works

- 1) Assign prior probabilities to each evolutionary scenario.

To calculate the GF prior probability, Aphid needs to calculate the frequency of discordant topologies:

$$f_{ac} = \frac{n_{tree}((AC)B)}{G}$$

$$f_{bc} = \frac{n_{tree}((BC)A)}{G}$$



How Aphid works

- 1) Assign prior probabilities to each evolutionary scenario.

To calculate the GF prior probability, Aphid needs to calculate the frequency of discordant topologies:

$$f_{ac} = \frac{n_{tree}((AC)B)}{G}$$

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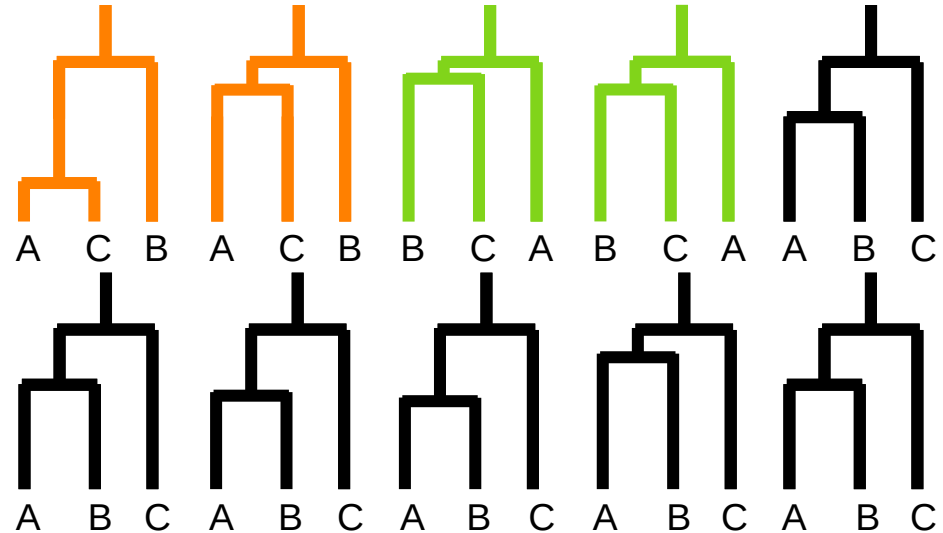
With the ILS probability, the GF probability can be calculated:

$$p_{ac} = f_{ac} - p_{ILS}/3$$

$$p_{bc} = f_{bc} - p_{ILS}/3$$

$$p_{ab} = \frac{p_{ac} - p_{bc}}{2}$$

$$P_{GF} = p_{ab} + p_{ac} + p_{bc}$$



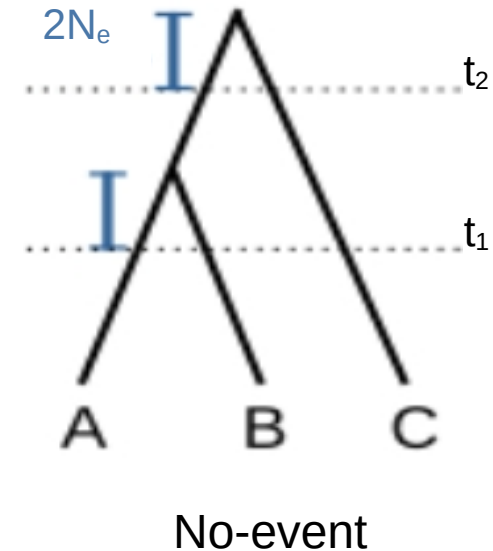
How Aphid works

1) Assign prior probabilities to each evolutionary scenario.

Calculate the probability specific to each scenario

Probability of observing a no-event scenario:

$$P(S_{\emptyset}) = (1 - p_{ILS}) \times (1 - p_{GF})$$



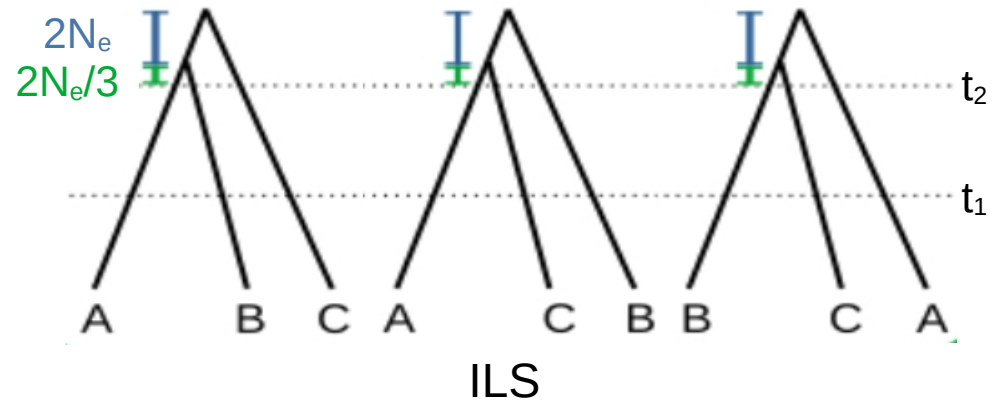
How Aphid works

1) Assign prior probabilities to each evolutionary scenario.

Calculate the probability specific for each scenarios

Probability of observing an ILS scenario:

$$P(S_{ILS(.)}) = \frac{1}{3} \times p_{ILS} \times (1 - p_{GF})$$



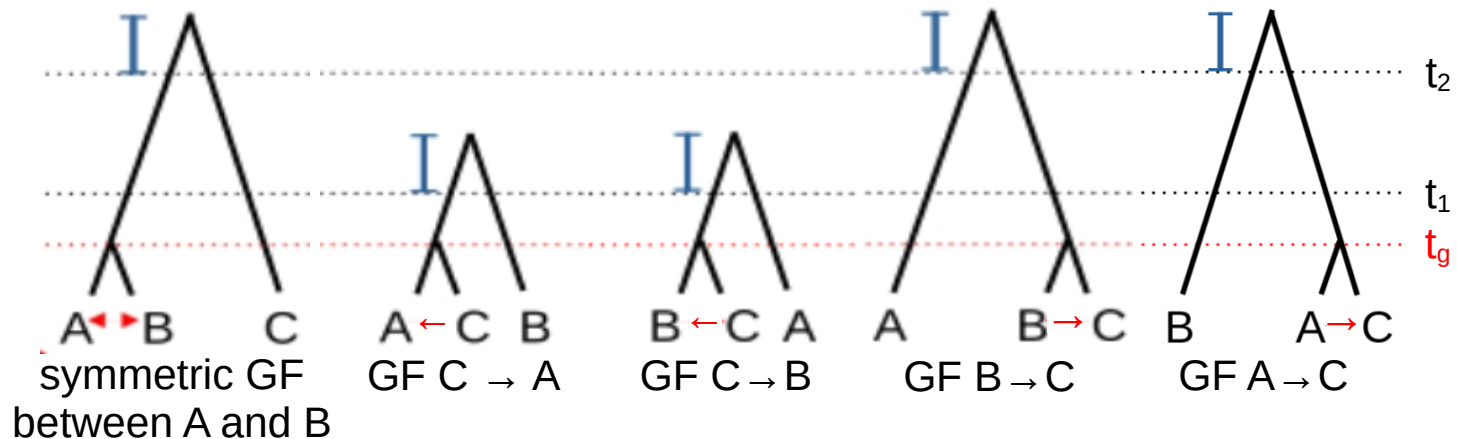
How Aphid works

1) Assign prior probabilities to each evolutionary scenario.

The number of GF scenarios depends on the number of GF timings tested.

=> Aphid needs to infer a probability per direction of GF and per timing tested.

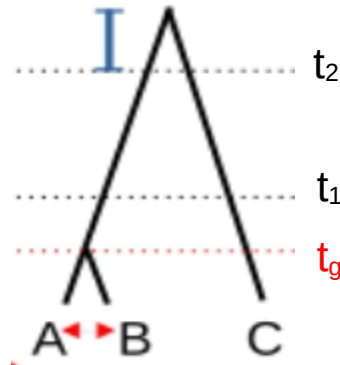
Probability that GF occurs at the later timing: $p_a = \frac{1}{|t_g|}$



How Aphid works

1) Assign prior probabilities to each evolutionary scenario.

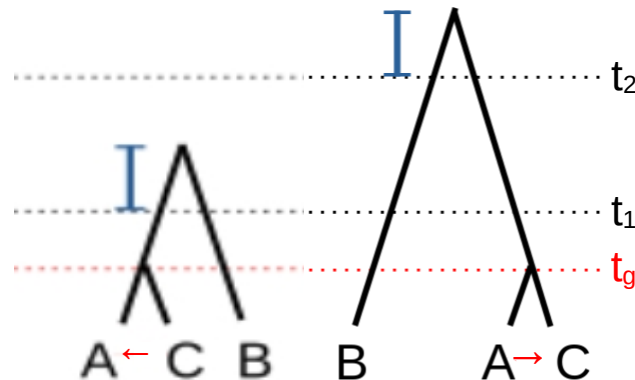
Direction	Ancestral timing	Other timings
$A \Leftrightarrow B$	$P(S_{GF(tg_i, ABC)}) = p_a \times p_{ab}$	$P(S_{GF(tg_i, ABC)}) = \frac{1 - p_a}{ tg - 1} \times p_{ab}$



How Aphid works

1) Assign prior probabilities to each evolutionary scenario.

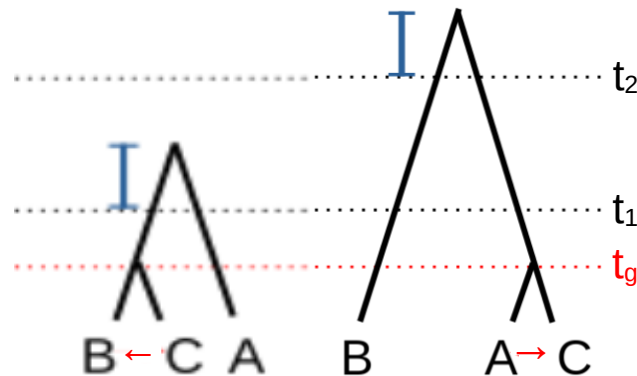
Direction	Ancestral timing	Other timings
$A \Leftrightarrow B$	$P(S_{GF(tg_i, ABC)}) = p_a \times p_{ab}$	$P(S_{GF(tg_i, ABC)}) = \frac{1 - p_a}{ tg - 1} \times p_{ab}$
$A \Rightarrow C$ or $C \Rightarrow A$	$P(S_{GF(tg_i, ACB)}) = p_a \times \frac{1}{2} \times p_{ac}$	$P(S_{GF(tg_i, ACB)}) = \frac{1 - p_a}{ tg - 1} \times \frac{1}{2} \times p_{ac}$



How Aphid works

1) Assign prior probabilities to each evolutionary scenario.

Direction	Ancestral timing	Other timings
$A \Leftrightarrow B$	$P(S_{GF(tg_i, ABC)}) = p_a \times p_{ab}$	$P(S_{GF(tg_i, ABC)}) = \frac{1 - p_a}{ tg - 1} \times p_{ab}$
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$B \Rightarrow C$ or $C \Rightarrow B$	$P(S_{GF(tg_i, BCA)}) = p_a \times \frac{1}{2} \times p_{bc}$	$P(S_{GF(tg_i, BCA)}) = \frac{1 - p_a}{ tg - 1} \times \frac{1}{2} \times p_{bc}$

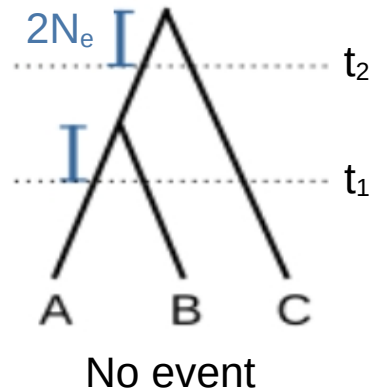


How Aphid works

2) Predict expected branch lengths for each scenario and likelihood.

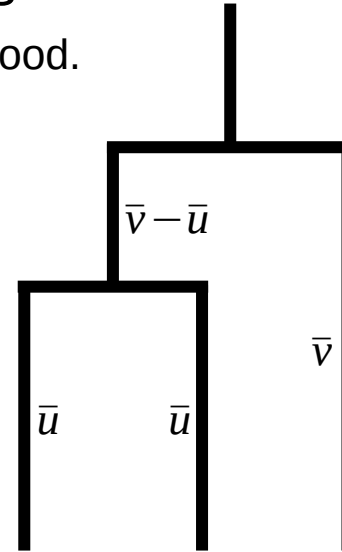
The branch lengths are defined as:

- $\bar{u}$, the short branches
- $\bar{v}$, the long branch
- $\bar{w} = \bar{v} - \bar{u}$, the internal branch length



$$\bar{u}(S_{\emptyset}) = \tau_1 + \frac{\theta}{2}$$

$$\bar{v}(S_{\emptyset}) = \tau_2 + \frac{\theta}{2}$$

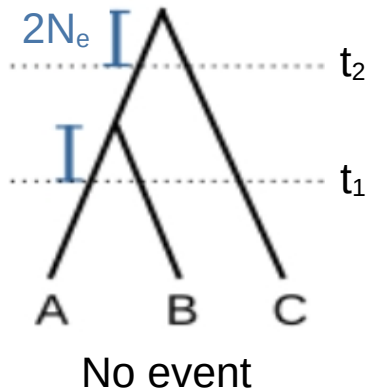


How Aphid works

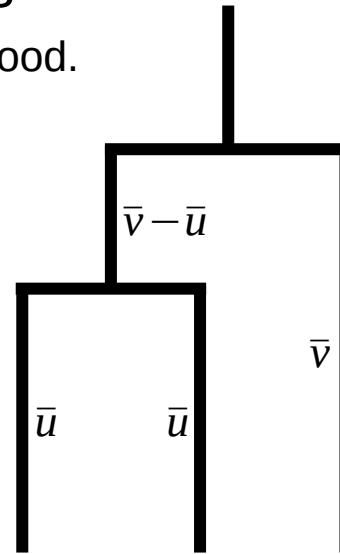
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$$\begin{aligned}\bar{u}(S_{\emptyset}) &= \tau_1 + \frac{\theta}{2} = \tau_1 + \frac{4 \times N_e \times \mu}{2} = (t_1 \times \mu) + (2 N_e \times \mu) \\ &= \mu \times (t_1 + 2 N_e) \\\bar{v}(S_{\emptyset}) &= \tau_2 + \frac{\theta}{2}\end{aligned}$$

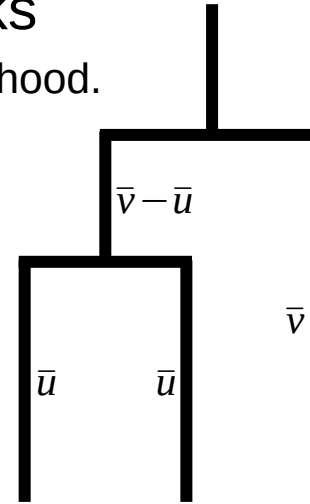


How Aphid works

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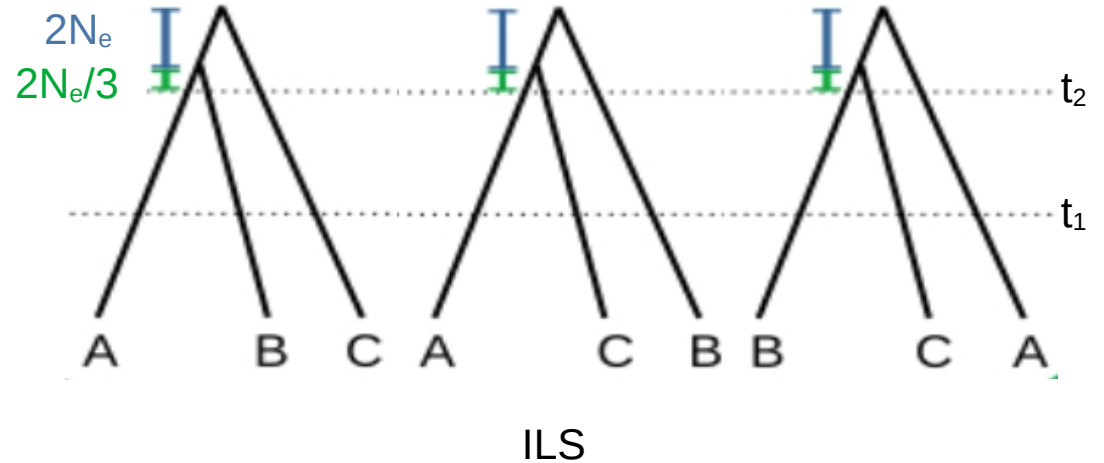
The branch lengths are defined as:

- $\bar{u}$, the short branches
- $\bar{v}$, the long branch
- $\bar{w} = \bar{v} - \bar{u}$, the internal branch length



$$\bar{u}(S_{ILS(.)}) = \tau_2 + \frac{\theta}{6}$$

$$\bar{v}(S_{ILS(.)}) = \tau_2 + \frac{2\theta}{3}$$

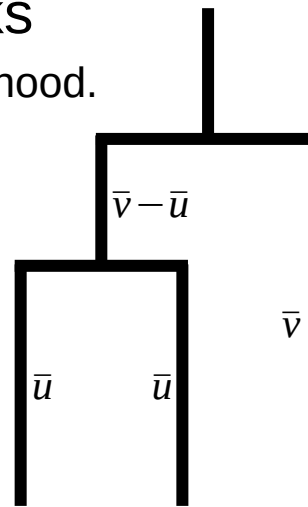


How Aphid works

2) Predict expected branch lengths for each scenario and likelihood.

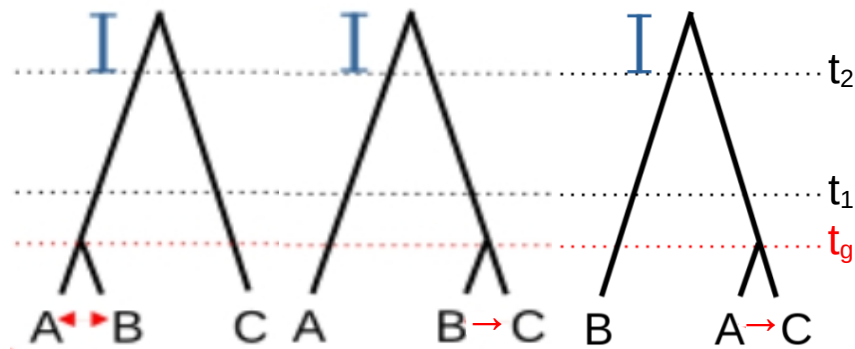
The branch lengths are defined as:

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$$\bar{u}(S_{GF(tg_i, \cdot)}) = tg_i$$

$$\bar{v}(S_{GF(\cdot, \{ABC, CAB, CBA\})}) = \tau_2 + \frac{\theta}{2}$$

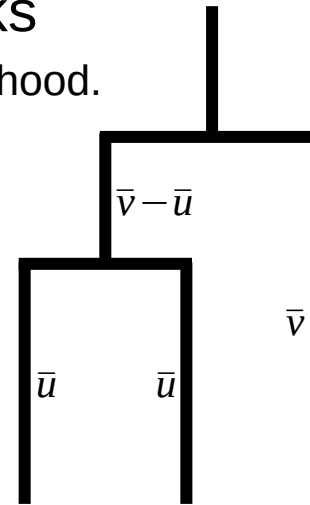


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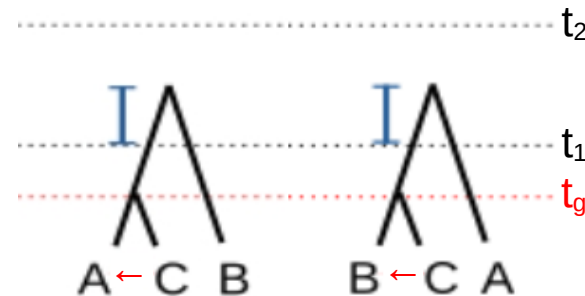
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$$\bar{v}(S_{GF(., \{ACB, BCA\})}) = \tau_1 + \frac{\theta}{2}$$



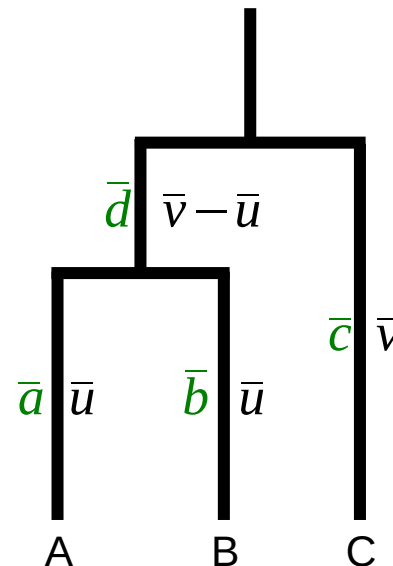
How Aphid works

2) Predict expected branch lengths for each scenario and likelihood.

The labels $\bar{u}$, $\bar{v}$, $\bar{w}$ are attributed to a triplet branch length following this table:

	T_g		
	ABC	ACB	BCA
$\bar{a}(s, T_g) =$	$\bar{u}(s)$	$\bar{u}(s)$	$\bar{v}(s)$
$\bar{b}(s, T_g) =$	$\bar{u}(s)$	$\bar{v}(s)$	$\bar{u}(s)$
$\bar{c}(s, T_g) =$	$\bar{v}(s)$	$\bar{u}(s)$	$\bar{u}(s)$

$$\bar{d}(s, \cdot) = \bar{v}(s) - \bar{u}(s)$$



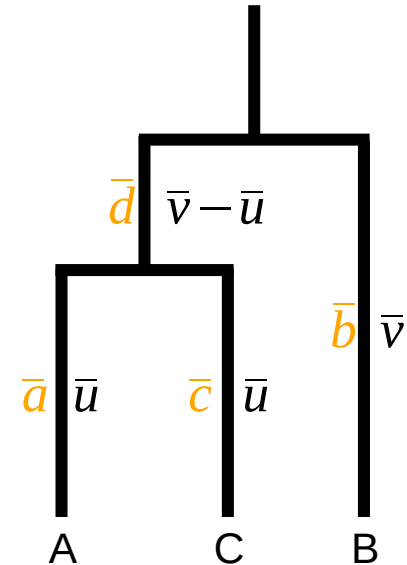
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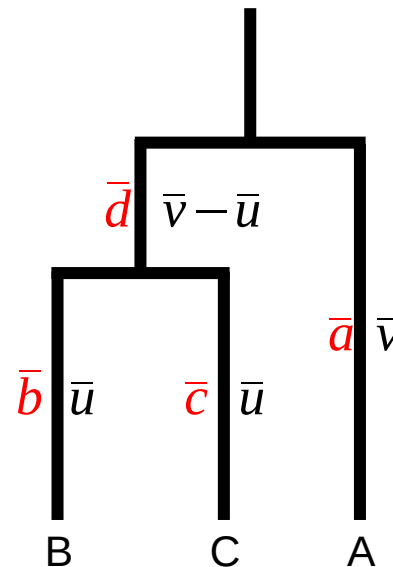
How Aphid works

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$$\bar{d}(s, \cdot) = \bar{v}(s) - \bar{u}(s)$$



How Aphid works

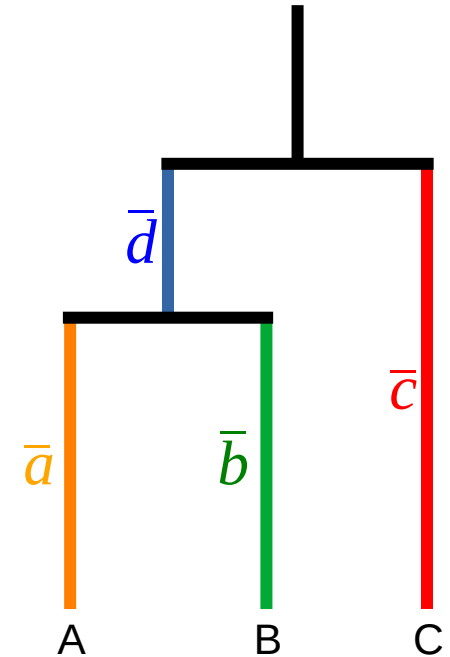
2) Predict expected branch lengths for each scenario and likelihood.

Calculate the likelihood of the expected length compared to the observation, using a Poisson probability model.

$$P(j, \lambda) = \frac{\lambda^j \times e^{-\lambda}}{j!}$$

The probability of a gene tree given a scenario is equal to:

$$P(G_i | S_k) = P(a_i, \alpha_i \times \bar{a}) \times P(b_i, \alpha_i \times \bar{b}) \times P(d_i, \alpha_i \times \bar{d}) \times P(c_i, \alpha_i \times \bar{c})$$



How Aphid works

2) Predict expected branch lengths for each scenario and likelihood.

The total probability of observing tree G_i is then calculated:

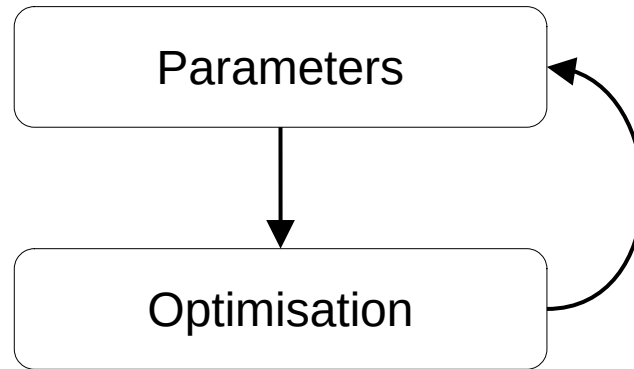
$$P(G_i) = \sum_k P(S_k) \times P(G_i | S_k)$$

Finally, the total probability of observing the data is:

$$\mathcal{L} = \prod_{G_i} P(G_i)$$

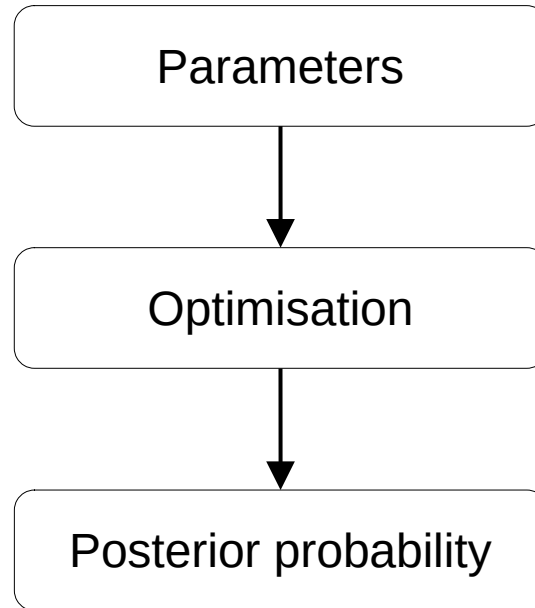
How Aphid works

3) Optimise the likelihood and the model's parameters.



How Aphid works

3) Optimise the likelihood and the model's parameters.



How Aphid works

4) Calculate the posterior probabilities of each scenario per gene.

The posterior probability of each of the scenarios for each gene is calculated as:

$$P(S_k|G_i) = \frac{P(S_k) \times P(G_i|S_k)}{P(G_i)}$$

How Aphid works

4) Calculate the posterior probabilities of each scenario per gene.

The posterior probability of each of the scenarios for each gene is calculated as:

$$P(S_k|G_i) = \frac{P(S_k) \times P(G_i|S_k)}{P(G_i)}$$

The posterior probabilities for a gene to be affected by ILS or GF can be calculated as:

$$P(ILS|G_i) = \sum_{S_k \in S_{ILS}} P(S_k|G_i)$$

$$P(GF|G_i) = \sum_{S_k \in S_{GF}} P(S_k|G_i)$$

How Aphid works

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The posterior probabilities for a gene to be affected by ILS or GF can be calculated as:

$$P(ILS|G_i) = \sum_{S_k \in S_{ILS}} P(S_k|G_i)$$

$$P(GF|G_i) = \sum_{S_k \in S_{GF}} P(S_k|G_i)$$

5) Average posterior probabilities to estimate the contribution of ILS and GF to the genetic conflict.

$$P_{ILS} = \frac{\sum_{T_i \neq ((A,B),C)} P(ILS|G_i)}{n}$$

$$P_{GF} = \frac{\sum_{T_i \neq ((A,B),C)} P(GF|G_i)}{n}$$

How Aphid works

6) Measure the imbalance between discordant topologies, separately for ILS and GF.

$$P_{\text{ILS}}[1] = \sum_{T_i=((A,C),B)} P(\text{ILS} | G_i) / n$$

$$P_{\text{ILS}}[2] = \sum_{T_i=((B,C),A)} P(\text{ILS} | G_i) / n$$

$$I_{\text{ILS}} = \frac{\max(P_{\text{ILS}}[1], P_{\text{ILS}}[2])}{P_{\text{ILS}}}$$

How Aphid works

6) Measure the imbalance between discordant topologies, separately for ILS and GF.

$$P_{\text{GF}}[1] = \sum_{T_i=((A,C),B)} P(\text{GF}|G_i)/n$$

$$P_{\text{GF}}[2] = \sum_{T_i=((B,C),A)} P(\text{GF}|G_i)/n$$

$$I_{\text{GF}} = \frac{\max(P_{\text{GF}}[1], P_{\text{GF}}[2])}{P_{\text{GF}}}$$

Aphid limitations

Aphid will provide:

- The proportions of genes affected by:
 - ILS
 - GF
 - Neither
- The discordant topology imbalance

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- The proportions of genes affected by:
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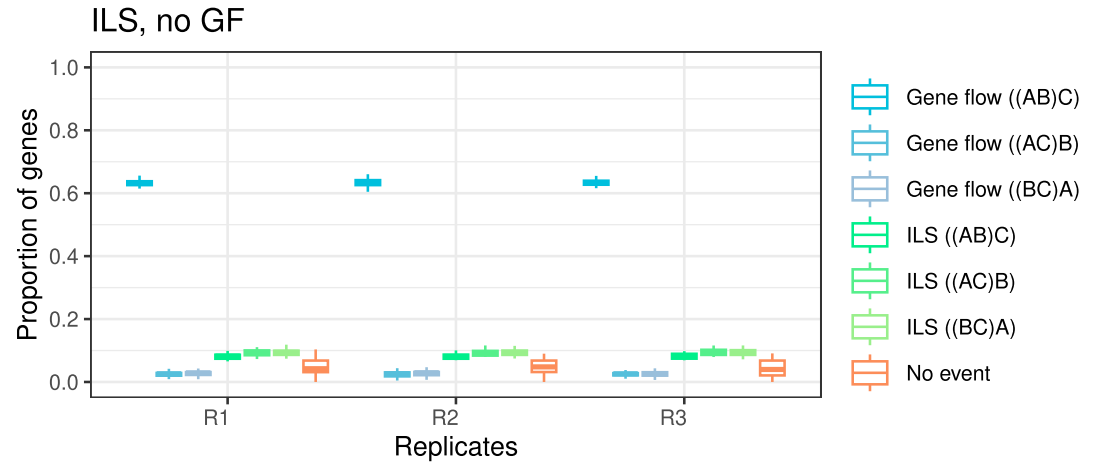
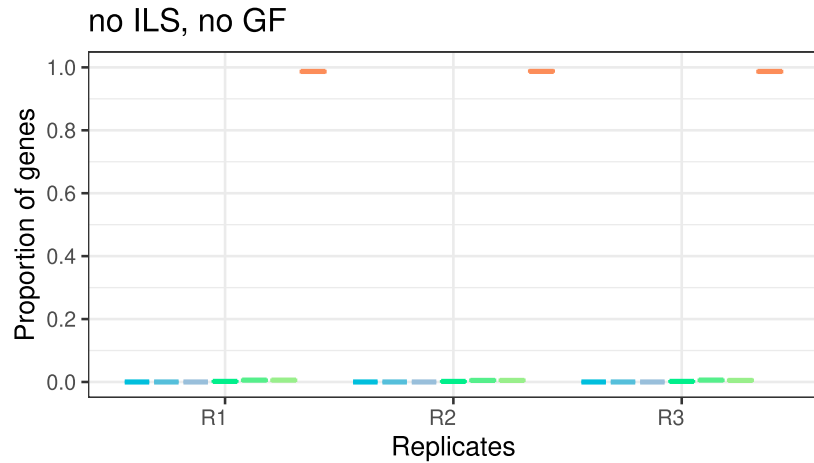
Aphid results can be used to:

- Distinguish the proportions between both discordant GF scenarios
- Calculate a median timing of GF

Aphid limitations

Aphid limitations:

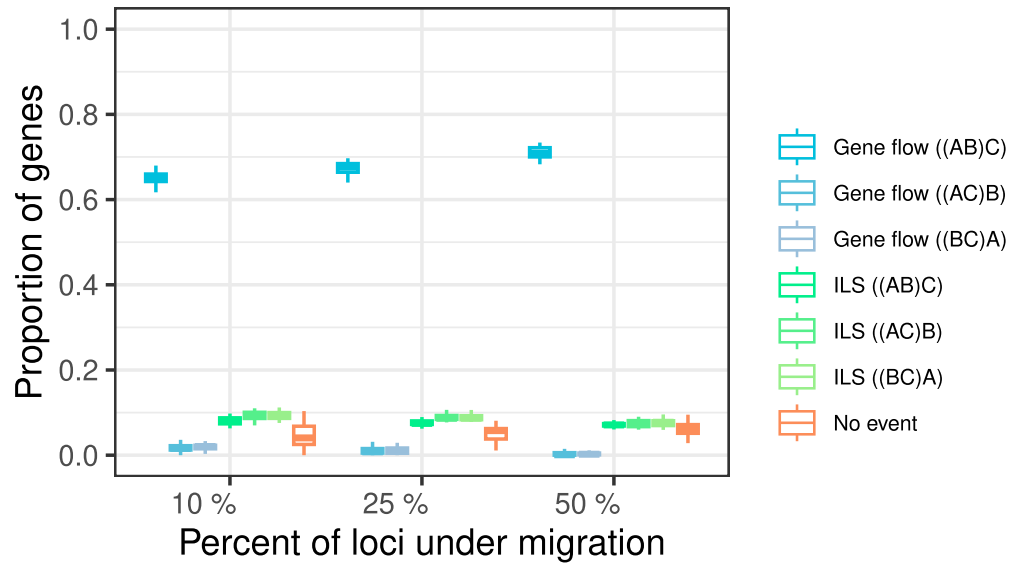
- Aphid cannot infer GF between sister species



Aphid limitations

Aphid **limitations**:

- Aphid cannot infer GF between sister species



Aphid limitations

Aphid **limitations**:

- Cannot infer GF between sister species
- The direction of GF cannot be inferred

